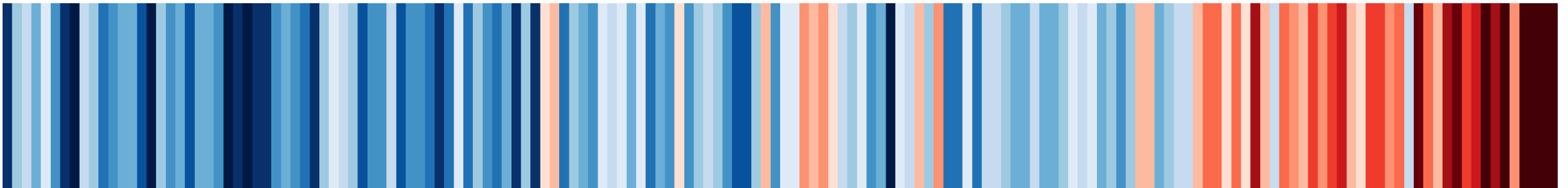


Psychology of Climate Change Communication

Week 2: Public opinion on climate change and the deficit model

Dr. Viktoria Cologna and Dr. Jan Pfänder



Let's recap

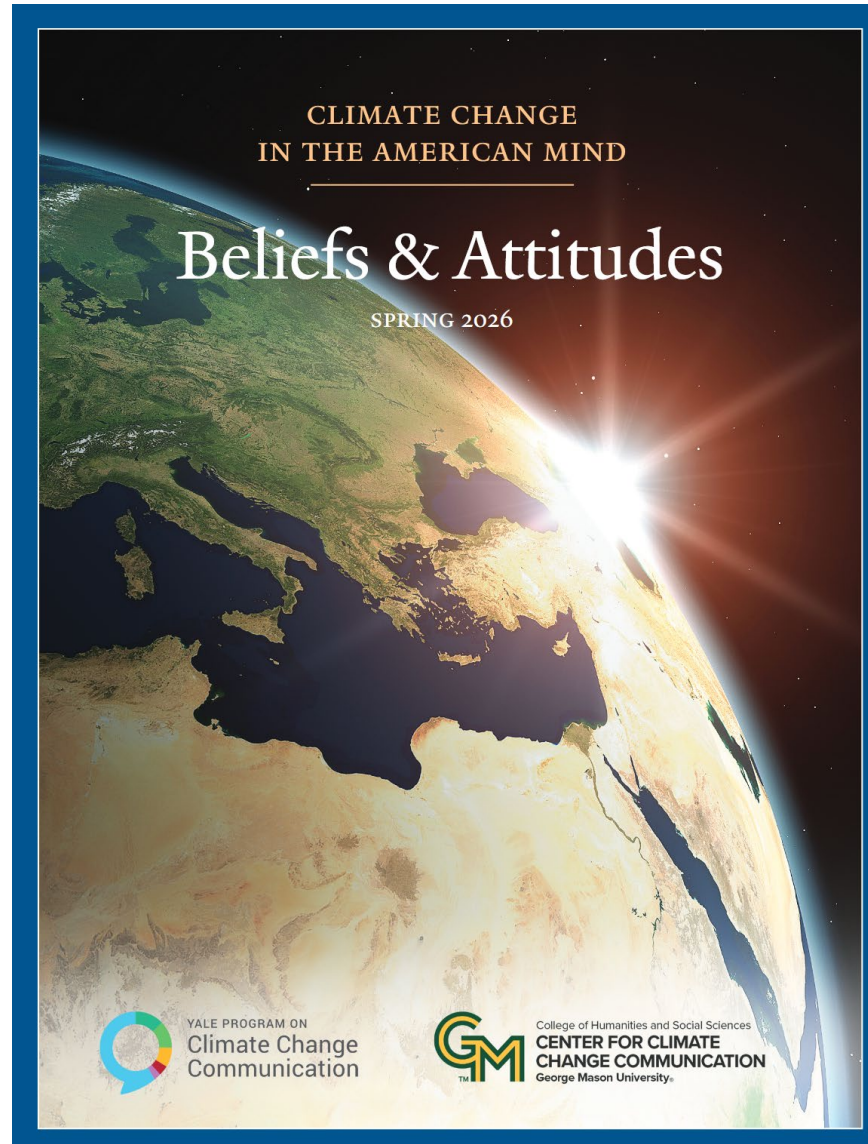
- Climate change communication is important:
 - It influences how individuals, organizations, and societies perceive climate change, and how they react to it.
- Climate change communication is challenging:
 - Not an “in-your-face-problem”
 - Psychological distance
 - Cognitive biases (e.g., temporal discounting, availability heuristic, confirmation bias)
 - Cognitive dissonance and motivated reasoning

Mentimeter

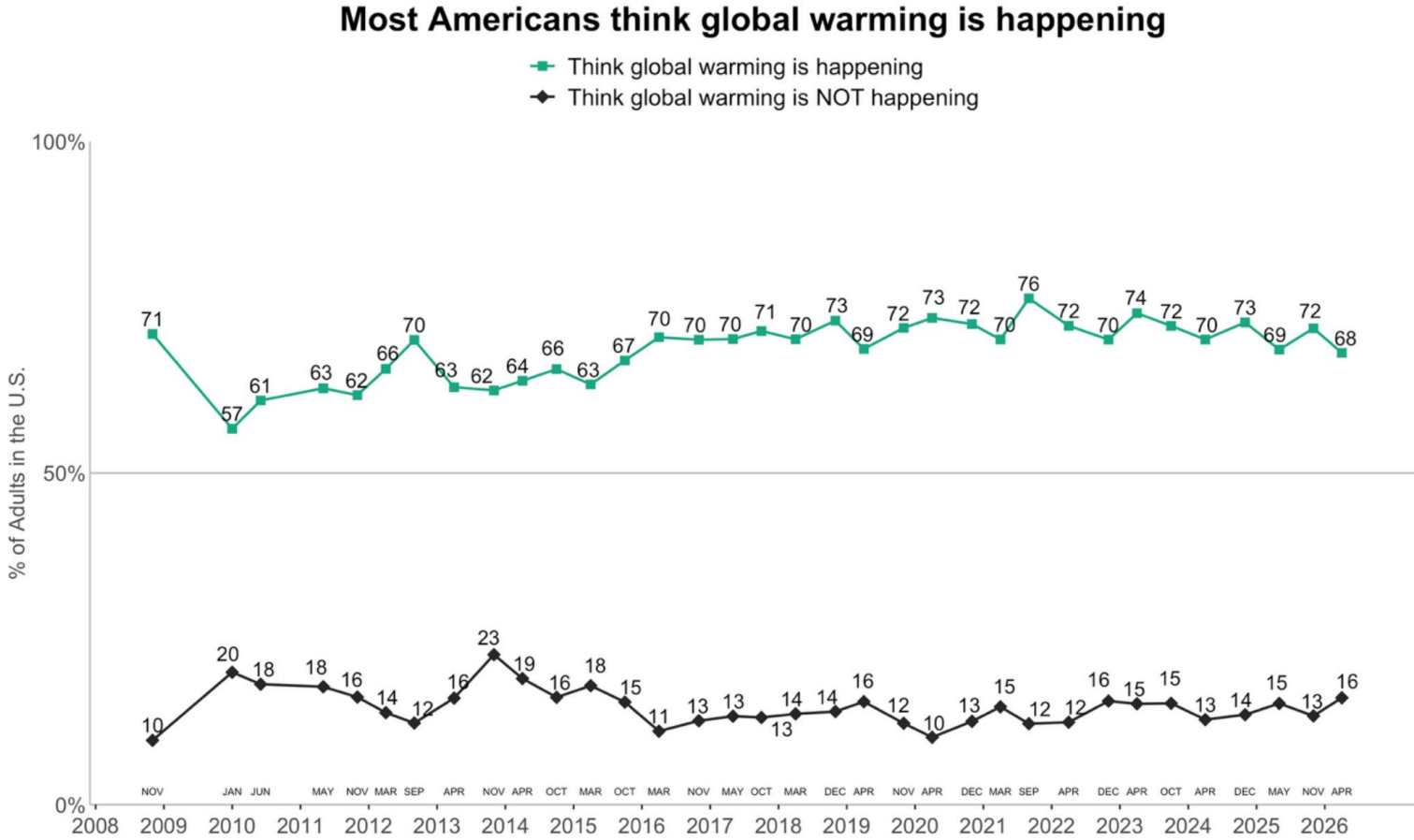
Today's lecture

- State of public opinion on climate change
- Communicating the scientific consensus
- Addressing different audiences
- Pluralistic ignorance
- The deficit model
- The role of knowledge in climate change communication

Public opinion about climate change – USA



Public opinion about climate change – USA



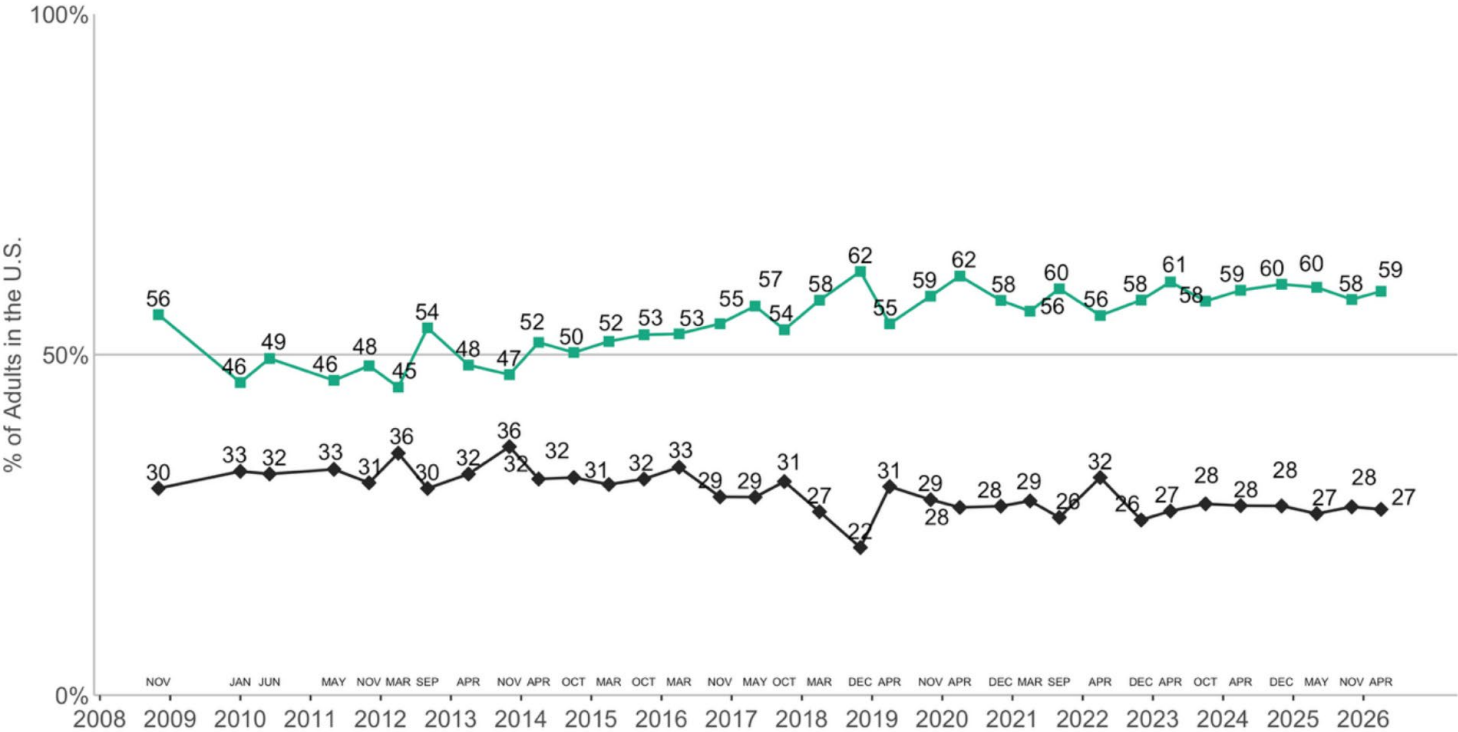
Do you think global warming is happening?

Adults in the U.S., Spring 2026

Public opinion about climate change – USA

A majority of Americans think global warming is mostly human-caused

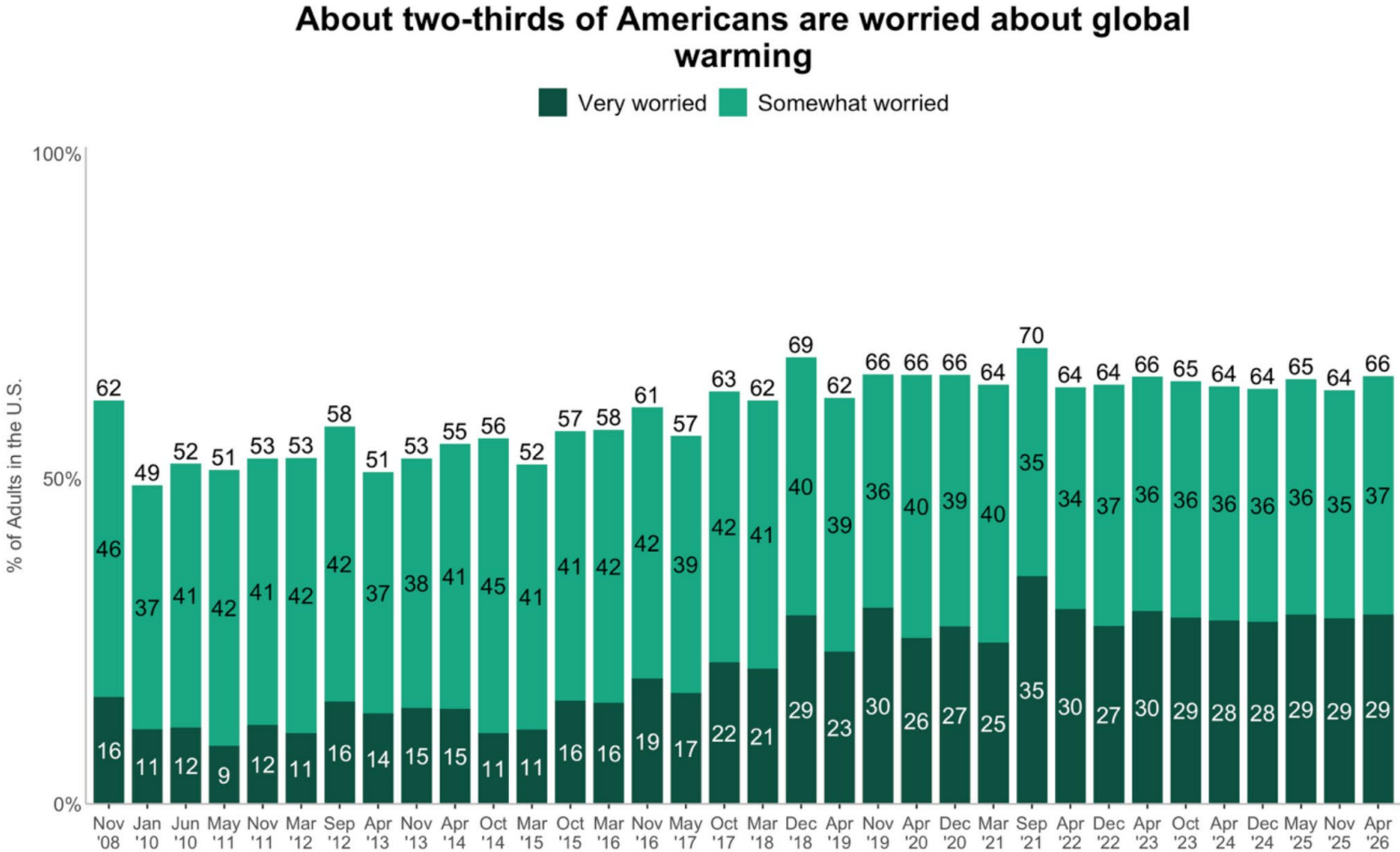
- Think global warming is caused mostly by human activities
- Think global warming is caused mostly by natural changes in the environment



Assuming global warming is happening, do you think it is . . .

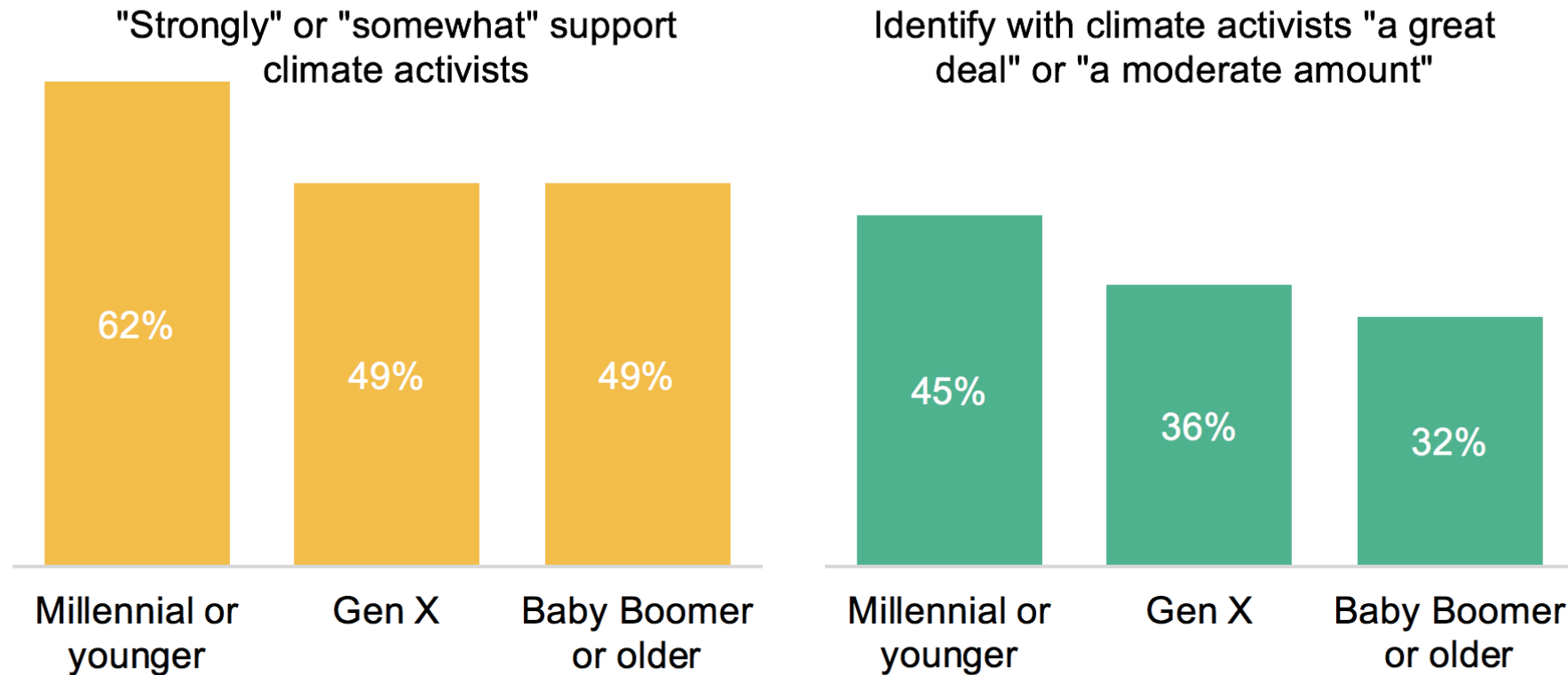
Adults in the U.S., Spring 2026

Public opinion about climate change – USA



How worried are you about global warming?

Younger generations of Americans, more than older generations, support and identify with climate activists who urge elected officials to act on global warming



Nov 2019. Base: 1,303 U.S. adults – Millennial or younger (1981 or later) $n = 375$; Gen X (1965 – 1980) $n = 336$; Baby Boomer or older (1964 or earlier) $n = 592$



YALE PROGRAM ON
Climate Change
Communication

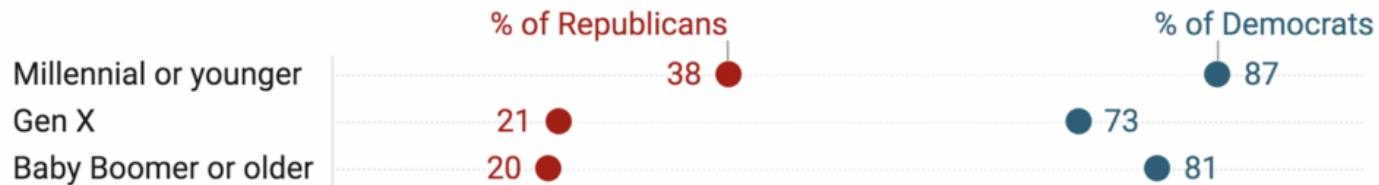


GEORGE MASON UNIVERSITY
CENTER for CLIMATE CHANGE
COMMUNICATION

Public opinion about climate change – USA

Young Republicans are more likely than older Republicans to support and identify with climate activists, and say they are willing to join a political campaign

Support climate activists who urge elected officials to take action to reduce global warming



Identify with climate activists who urge elected officials to take action to reduce global warming



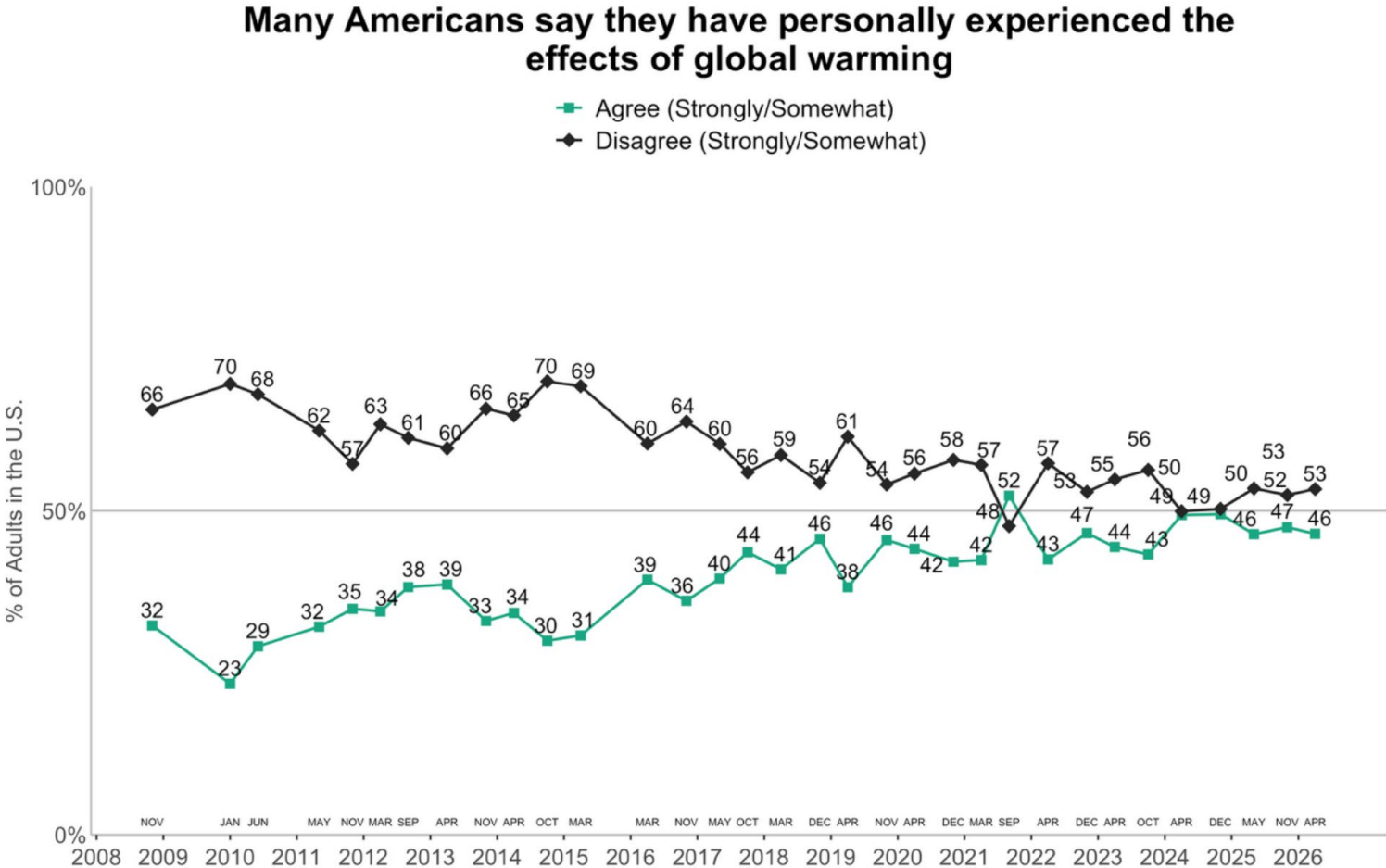
Willing to join a campaign to convince elected officials to take action to reduce global warming



% who "strongly" or "somewhat" support climate activists; identify with climate activists "a great deal" or "a moderate amount," and would "definitely" or "probably" join a political campaign

Nov 2019. Base: 1,045 U.S. adults – Millennial or younger (1981 or later) $n = 283$; Gen X (1965 – 1980) $n = 262$; Baby Boomer or older (1964 or earlier) $n = 500$

Public opinion about climate change – USA

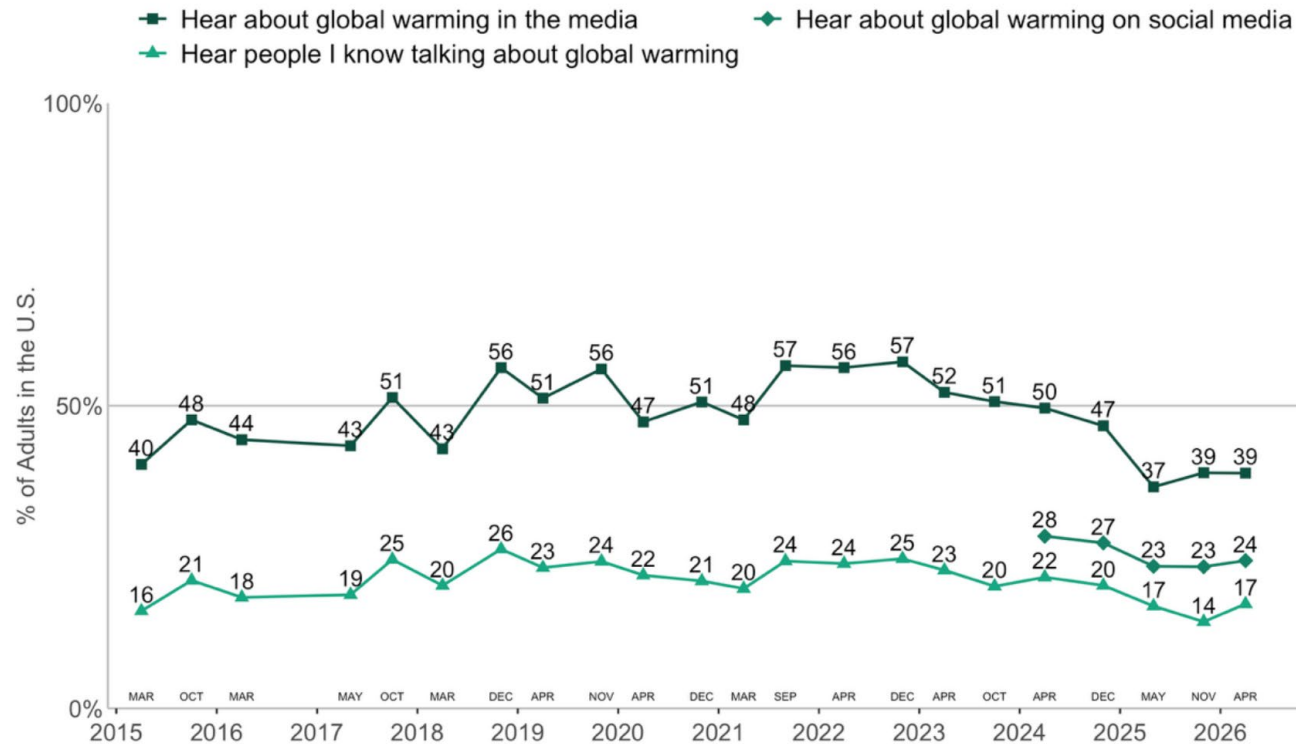


How much do you agree or disagree with the following statement? I have personally experienced the effects of global warming.

Adults in the U.S., Spring 2026

Public opinion about climate change – USA

The proportion of Americans who hear about global warming has decreased in recent years



About how often do you hear about global warming in the media (TV, movies, radio, newspapers/news websites, magazines, etc.)?

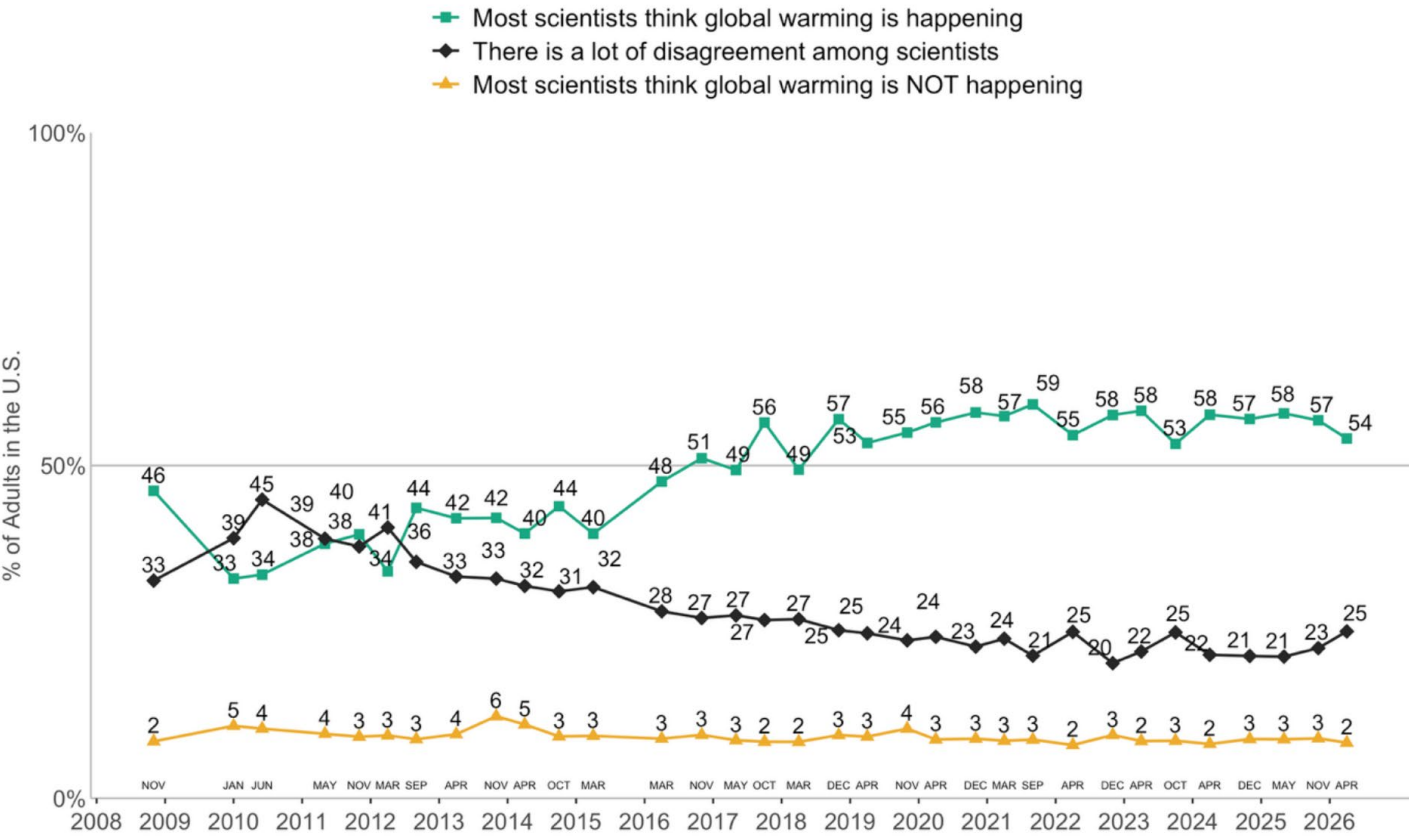
About how often do you hear about global warming on social media (Facebook, Instagram, TikTok, YouTube, Twitter/X, etc.)?

About how often do you hear other people you know (your family, friends, co-workers, etc.) talk about global warming?

Adults in the U.S., Spring 2026

Public opinion about climate change – USA

A majority of Americans understand that most scientists think global warming is happening

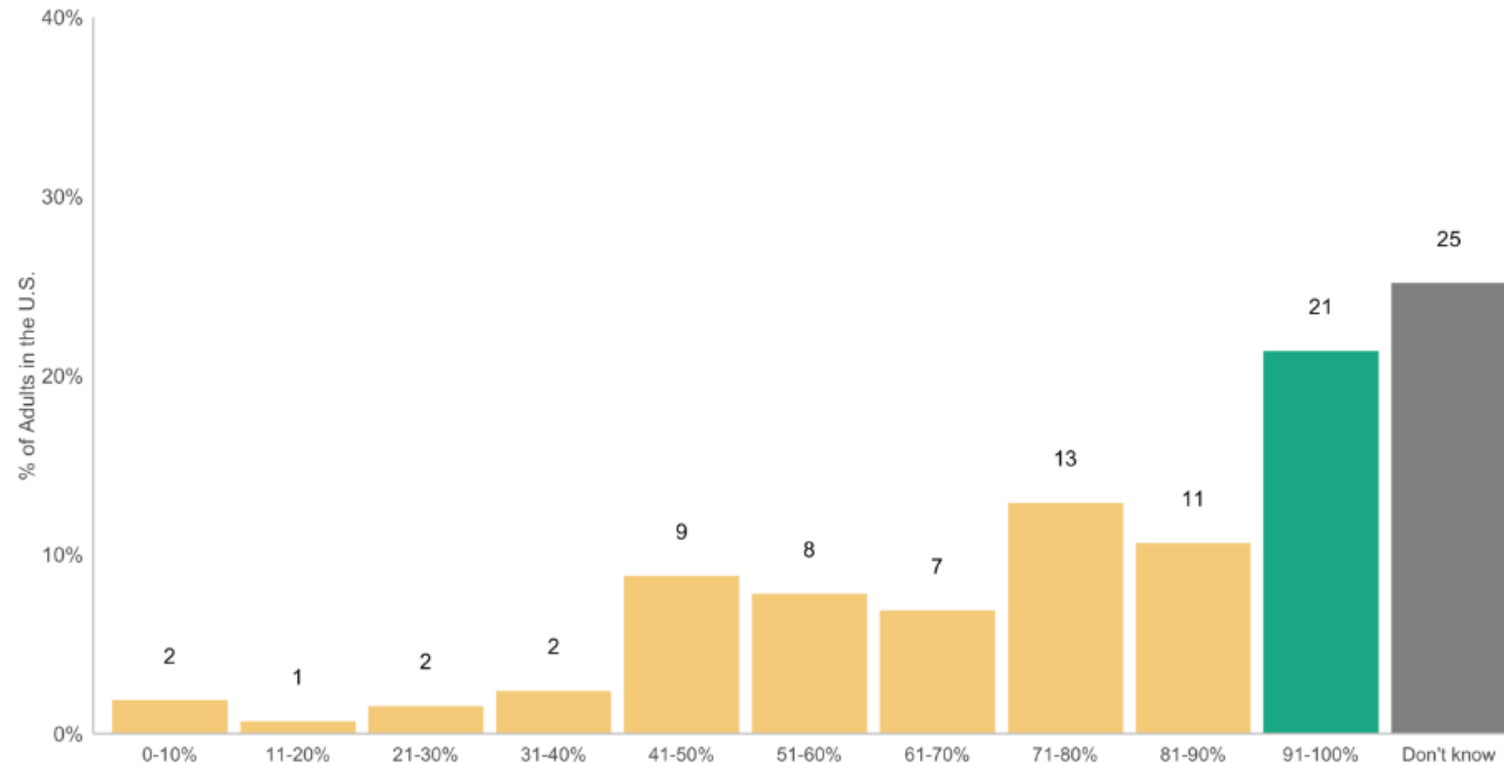


What comes closest to your own view?

Adults in the U.S., Spring 2026

Public opinion about climate change – USA

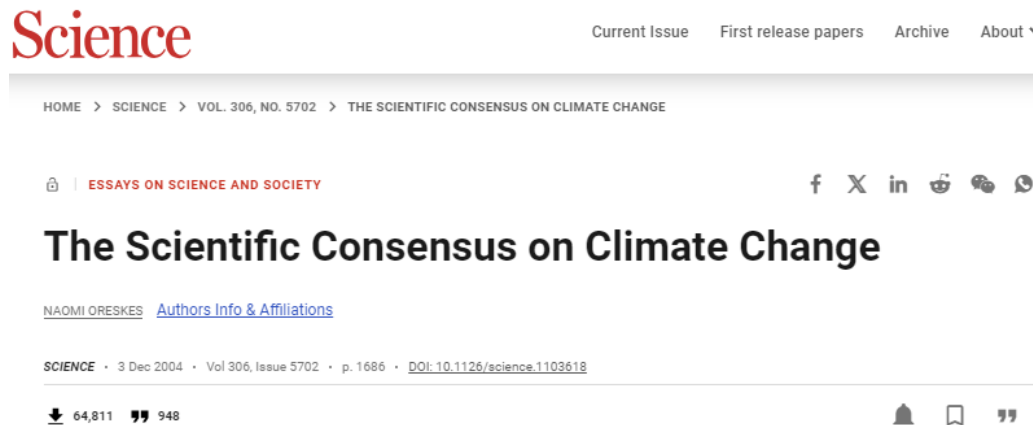
About one in five Americans understand that almost all climate scientists (more than 90%) have concluded that human-caused global warming is happening



To the best of your knowledge, what percentage of climate scientists think that human-caused global warming is happening?

Spring 2024

How do we know there is scientific consensus on climate change?



“That hypothesis was tested by analyzing 928 abstracts, published in refereed scientific journals between 1993 and 2003, and listed in the ISI database with the keywords “climate change”.



Naomi Oreskes, Harvard University



Bojana Većkalov

nature human behaviour



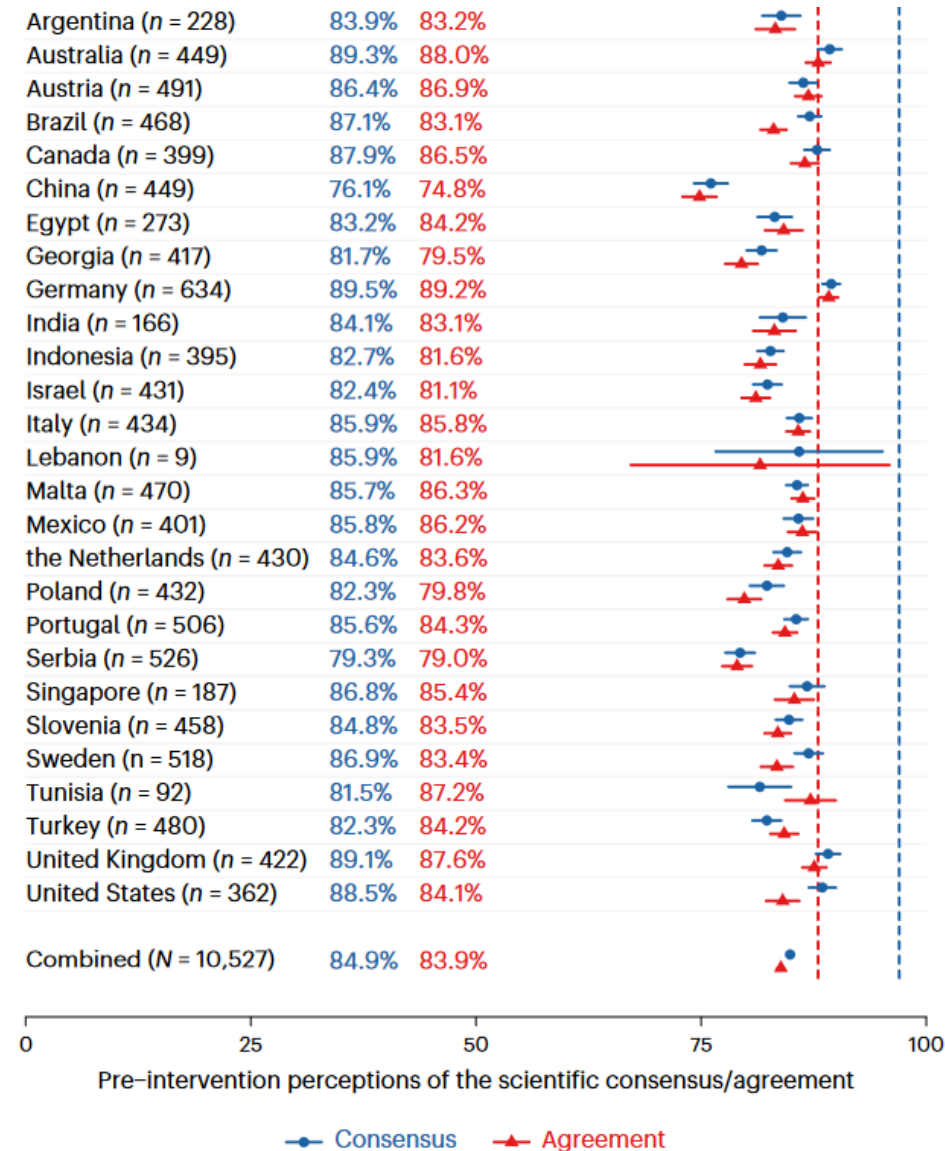
Article

<https://doi.org/10.1038/s41562-024-01928-2>

A 27-country test of communicating the scientific consensus on climate change

Misperceptions of the scientific consensus

Fig. 1 | Mean pre-intervention (mis)perceptions of the scientific consensus on the reality of climate change and agreement on climate change as a crisis per country sample. The error bars represent the 95% CI for each country. The dashed blue line represents the actual scientific reality consensus (97%). The dashed red line represents the actual scientific crisis agreement (88%).



Correcting misperceptions of the scientific consensus

Article

<https://doi.org/10.1038/s41562-024-01928-2>

A 27-country test of communicating the scientific consensus on climate change

Received: 21 August 2023

Accepted: 13 June 2024

Published online: 26 August 2024

 Check for updates

A list of authors and their affiliations appears at the end of the paper

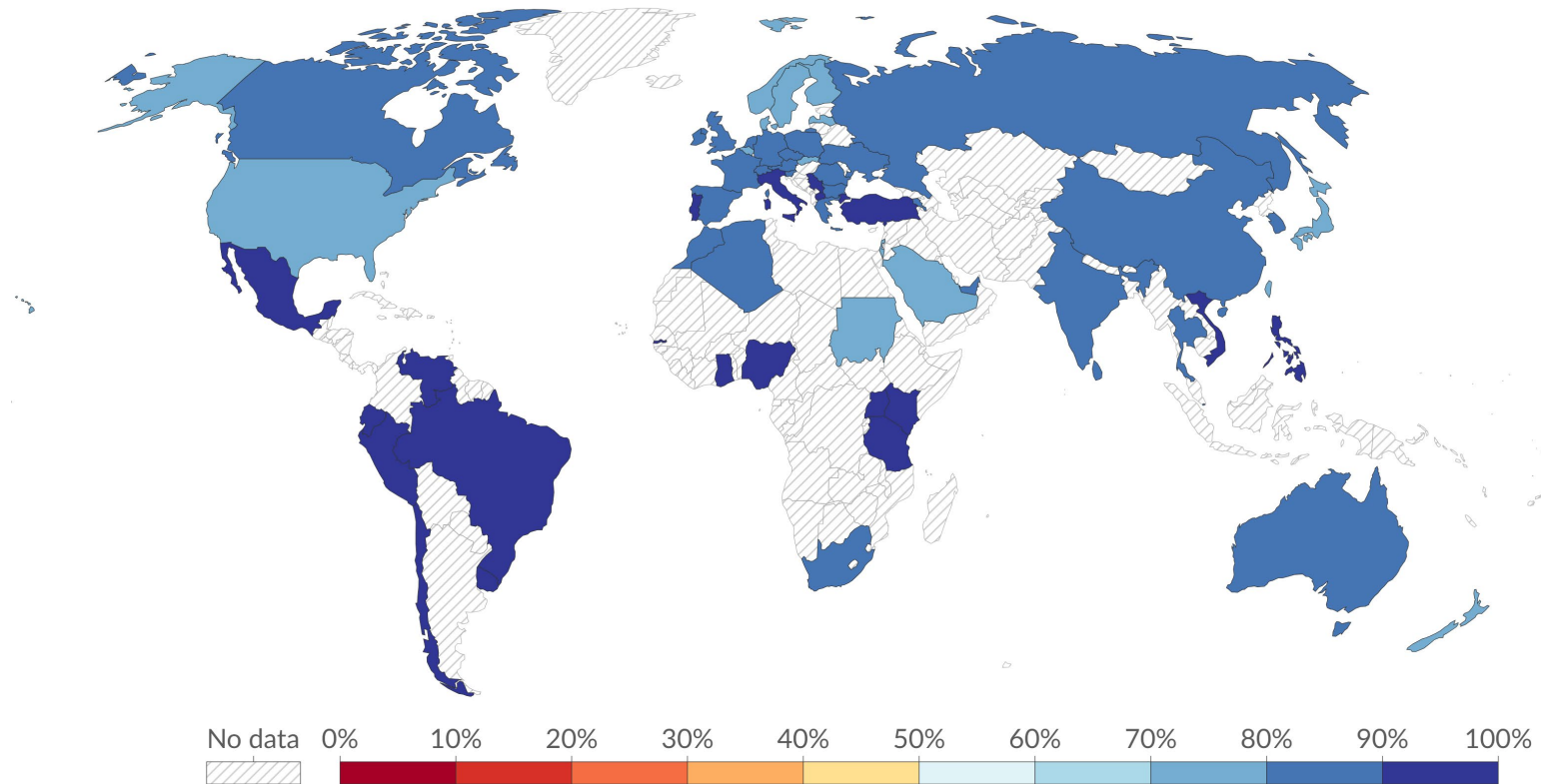
Communicating the scientific consensus that human-caused climate change is real increases climate change beliefs, worry and support for public action in the United States. In this preregistered experiment, we tested two scientific consensus messages, a classic message on the reality of human-caused climate change and an updated message additionally emphasizing scientific agreement that climate change is a crisis. Across online convenience samples from 27 countries ($n = 10,527$), the classic message substantially reduces misperceptions ($d = 0.47$, 95% CI (0.41, 0.52)) and slightly increases climate change beliefs (from $d = 0.06$, 95% CI (0.01, 0.11) to $d = 0.10$, 95% CI (0.04, 0.15)) and worry ($d = 0.05$, 95% CI (-0.01, 0.10)) but not support for public action directly. The updated message is equally effective but provides no added value. Both messages are more effective for audiences with lower message familiarity and higher misperceptions, including those with lower trust in climate scientists and right-leaning ideologies. Overall, scientific consensus messaging is an effective, non-polarizing tool for changing misperceptions, beliefs and worry across different audiences.

Public opinion about climate change – global

Share of people who believe in climate change and think it's a serious threat to humanity 2023

Our World
in Data

Participants were asked to score beliefs on a scale from 0 to 100 on four questions: whether action was necessary to avoid a global catastrophe; humans were causing climate change; it was a serious threat to humanity; and was a global emergency.



Data source: Vlasceanu et al. (2024). Addressing climate change with behavioral science: A global intervention tournament in 63 countries.

Note: Based on survey data across almost 60,000 participants from 63 countries.

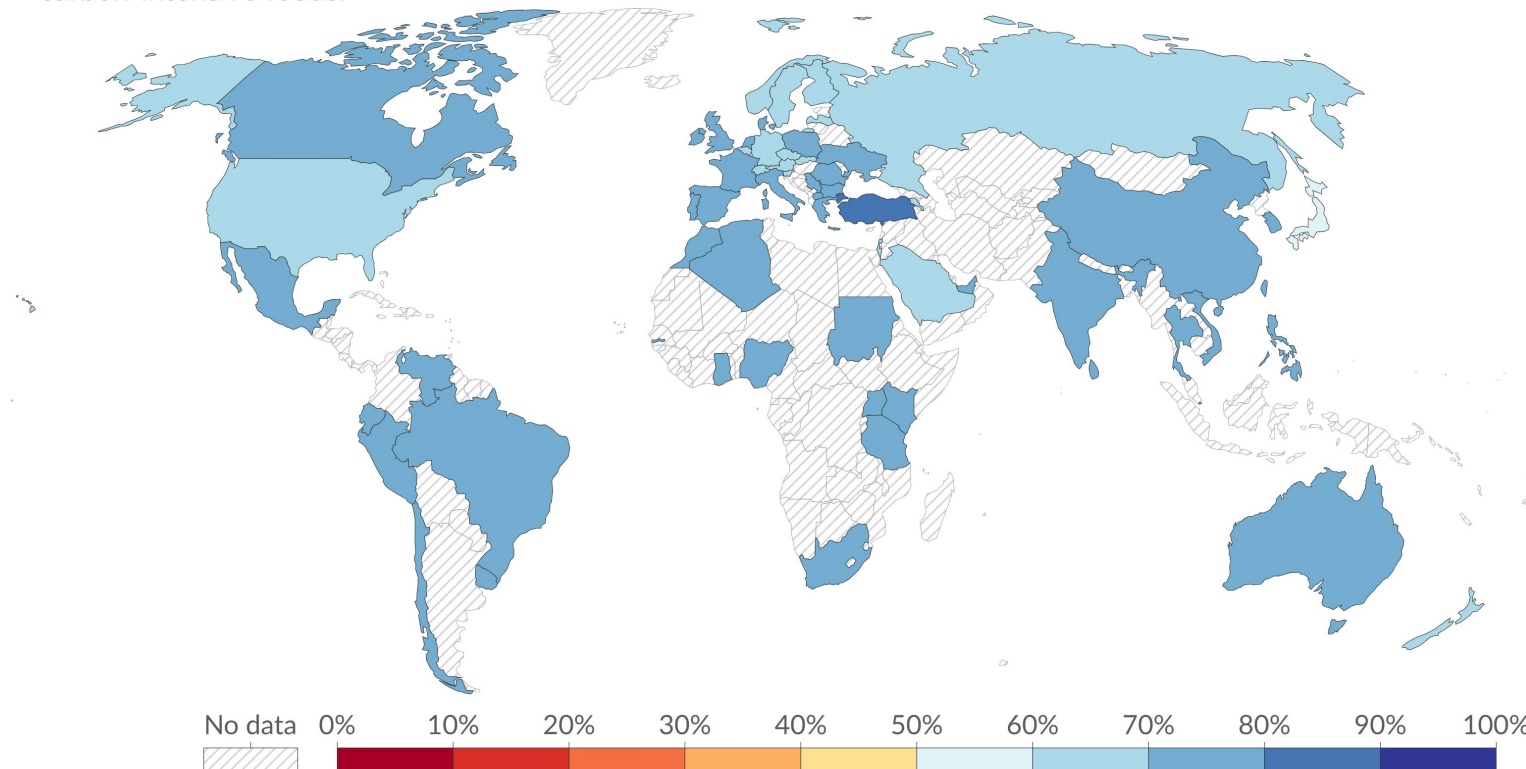
OurWorldinData.org/climate-change | CC BY

Public opinion about climate change – global

Share of people who support policies to tackle climate change 2023

Our World
in Data

Support was measured on a scale from 0 to 100 across nine interventions, including carbon taxes on fossil fuels, expanding public transport, more renewable energy, more electric car chargers, taxes on airlines, investments in green jobs and businesses, laws to keep waterways clean, protecting forests, and increasing taxes on carbon-intensive foods.

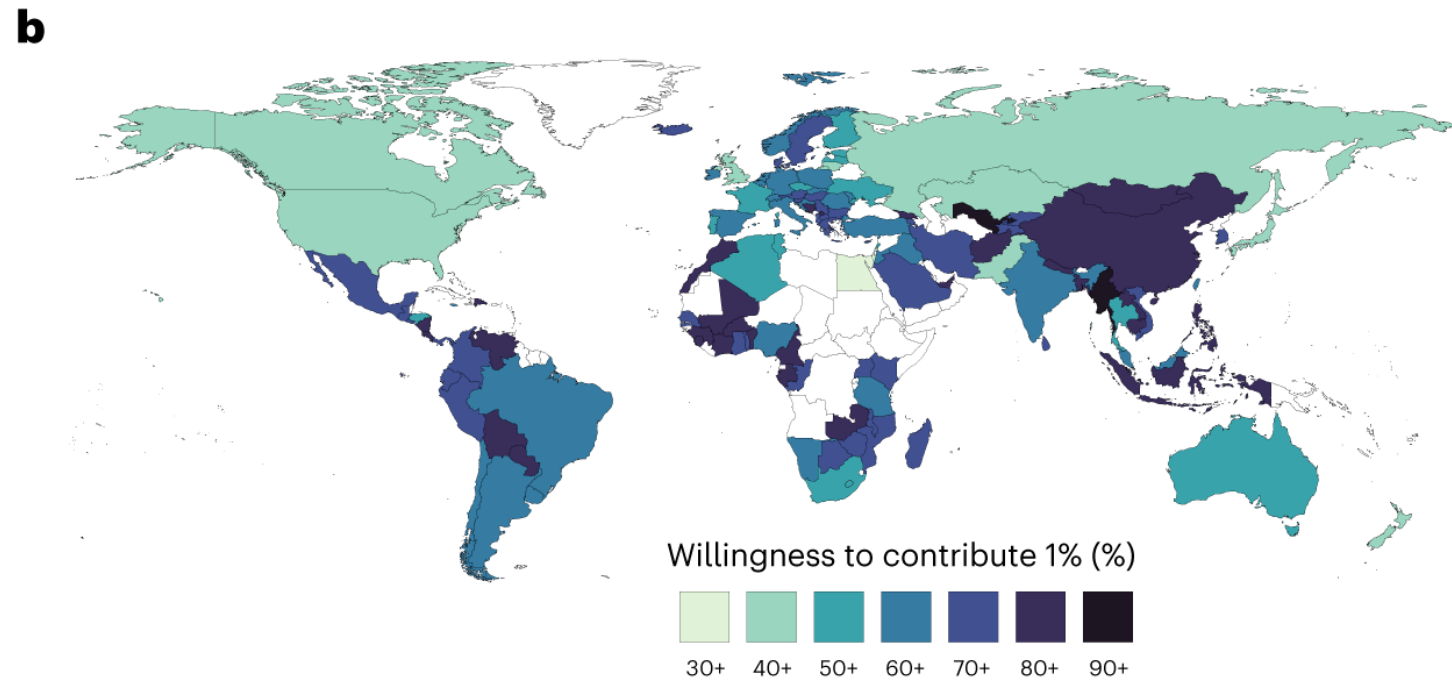
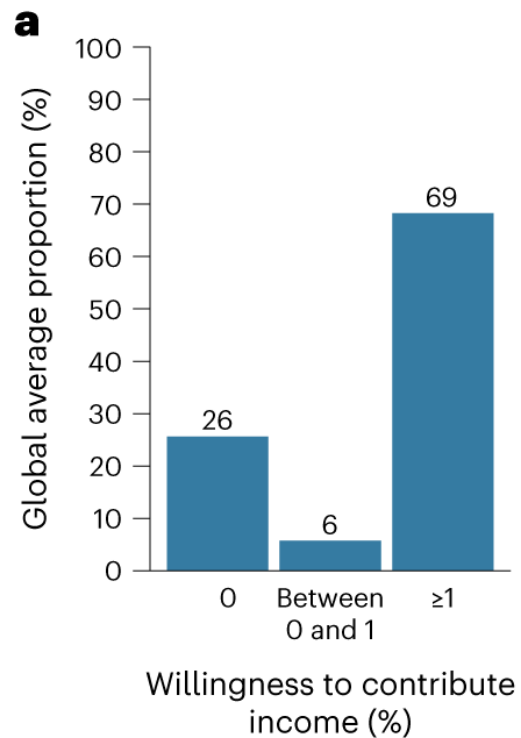


Data source: Vlasceanu et al. (2024). Addressing climate change with behavioral science: A global intervention tournament in 63 countries.

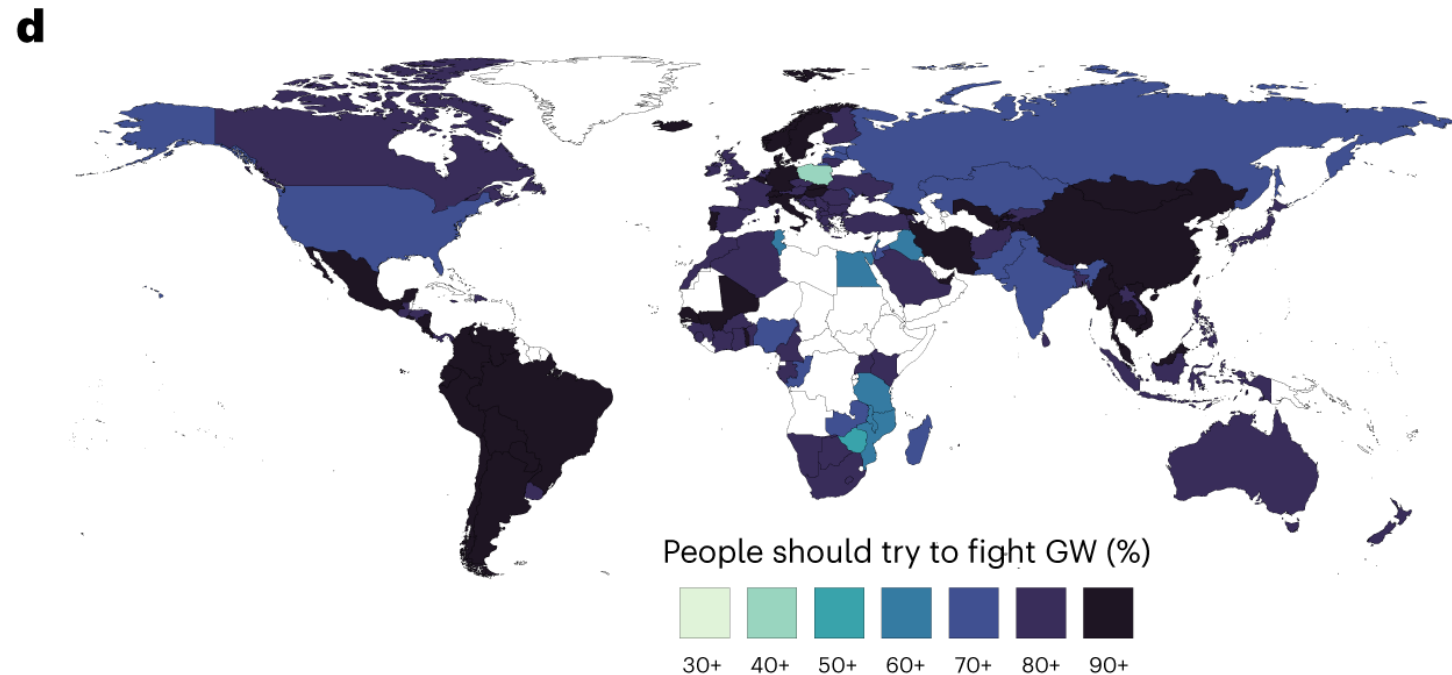
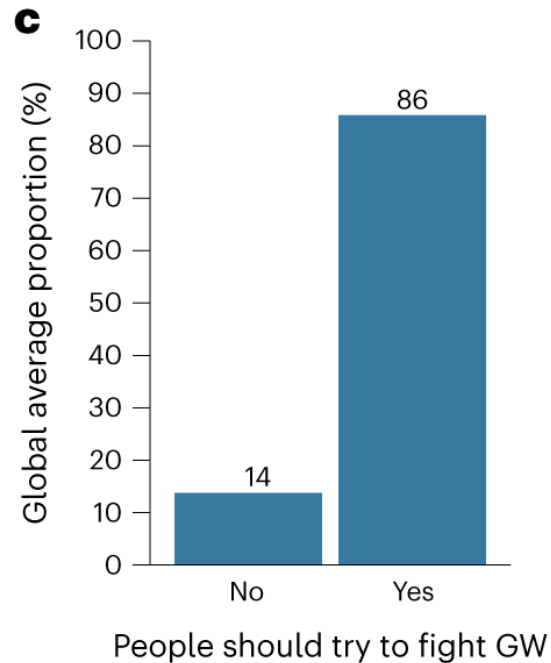
Note: The final score was based on the average of the nine interventions, each measured on a 0 to 100 scale.

OurWorldinData.org/climate-change | CC BY

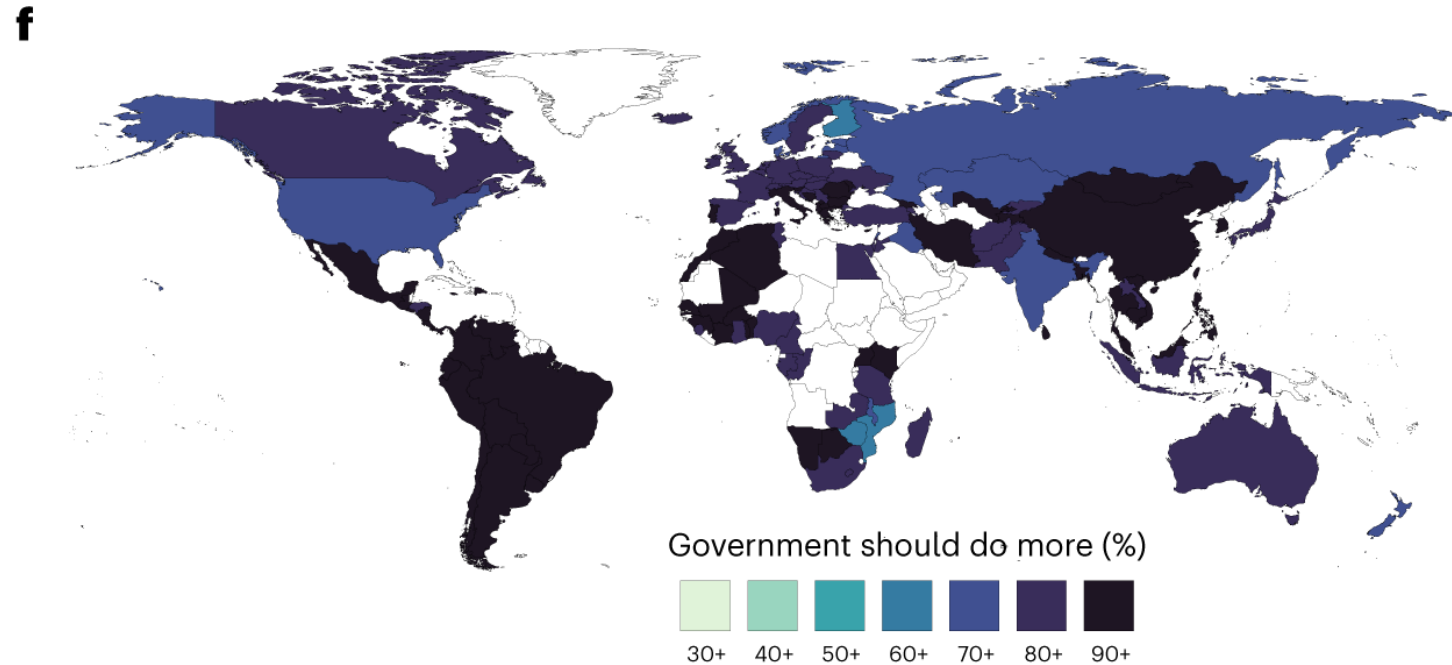
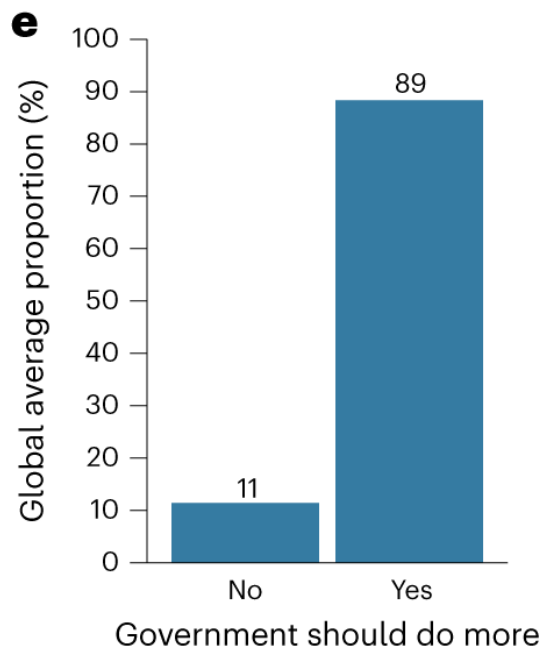
Willingness to contribute 1% income



People should try to fight global warming



Government should do more



So why do people not engage more if they support it?

People underestimate others' willingness to take climate action

Our World
in Data

Participants were asked if they would contribute 1% of their income to tackle climate change. The share that answered "yes" is shown on the horizontal axis. The share of the population in their country that people *think* would be willing is shown on the vertical axis.

Predicted share willing to give



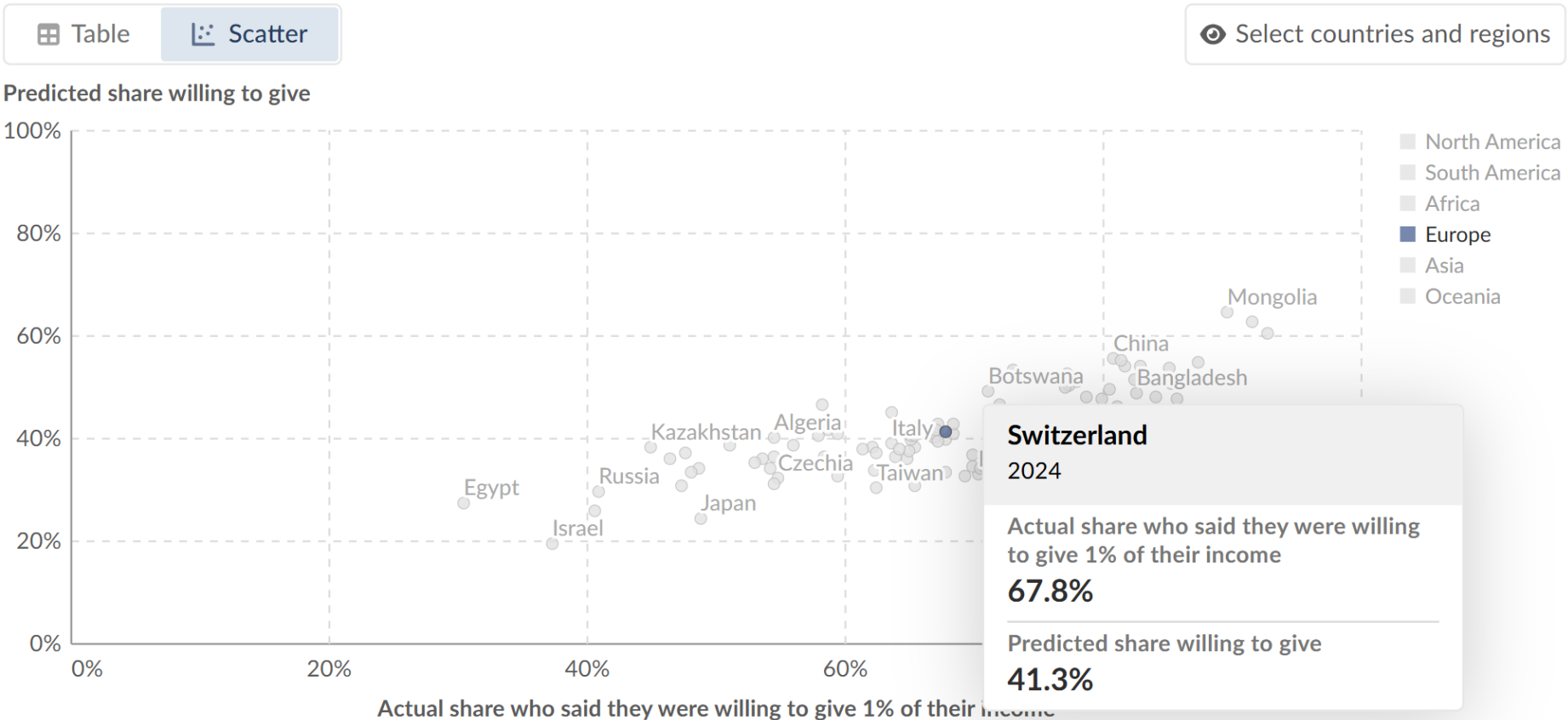
Data source: Andre et al. (2024). Globally representative evidence on the actual and perceived support for climate action.

Note: Based on representative surveys of almost 130,000 people across 125 countries.

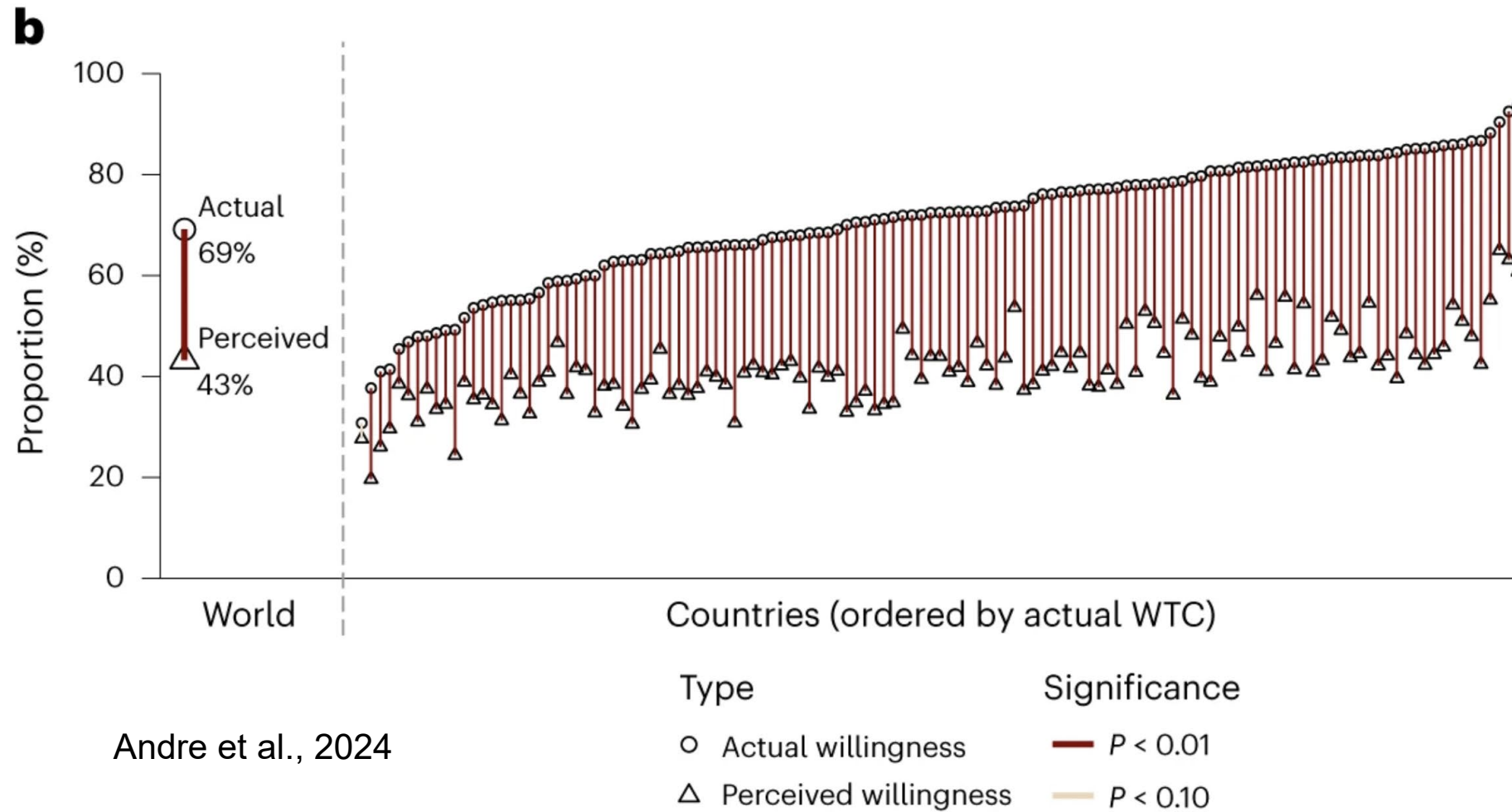
OurWorldinData.org/climate-change | CC BY

People underestimate others' willingness to take climate action

Participants were asked if they would contribute 1% of their income to tackle climate change. The share that answered "yes" is shown on the horizontal axis. The share of the population in their country that people *think* would be willing is shown on the vertical axis.



Same data, different visualization



Andre et al., 2024

Widespread evidence for pluralistic evidence on climate change

Article | [Open access](#) | Published: 23 August 2022

Americans experience a false social reality by underestimating popular climate policy support by nearly half

[Gregg Sparkman](#) , [Nathan Geiger](#) & [Elke U. Weber](#)

[Nature Communications](#) **13**, Article number: 4779 (2022) | [Cite this article](#)



- While 66–80% Americans support these policies, Americans estimate the prevalence to only be between 37–43% on average.
- Thus, supporters of climate policies outnumber opponents two to one, while Americans falsely perceive nearly the opposite to be true.

Mentimeter results

So why don't people engage more if they support engagement?

- One reason: There is a gap between what people believe and what they think that other people believe.
- **Pluralistic ignorance** is a phenomenon in which people mistakenly believe that others predominantly hold an opinion different from their own. (Moser, 2010)

Example:

College students generally believed they were more uncomfortable with campus alcohol practices than the average student → misperception. Over the course of a semester, male (but not female) students shifted their attitudes over time in the direction of what they mistakenly believed to be the norm (Prentice & Miller, 1993)

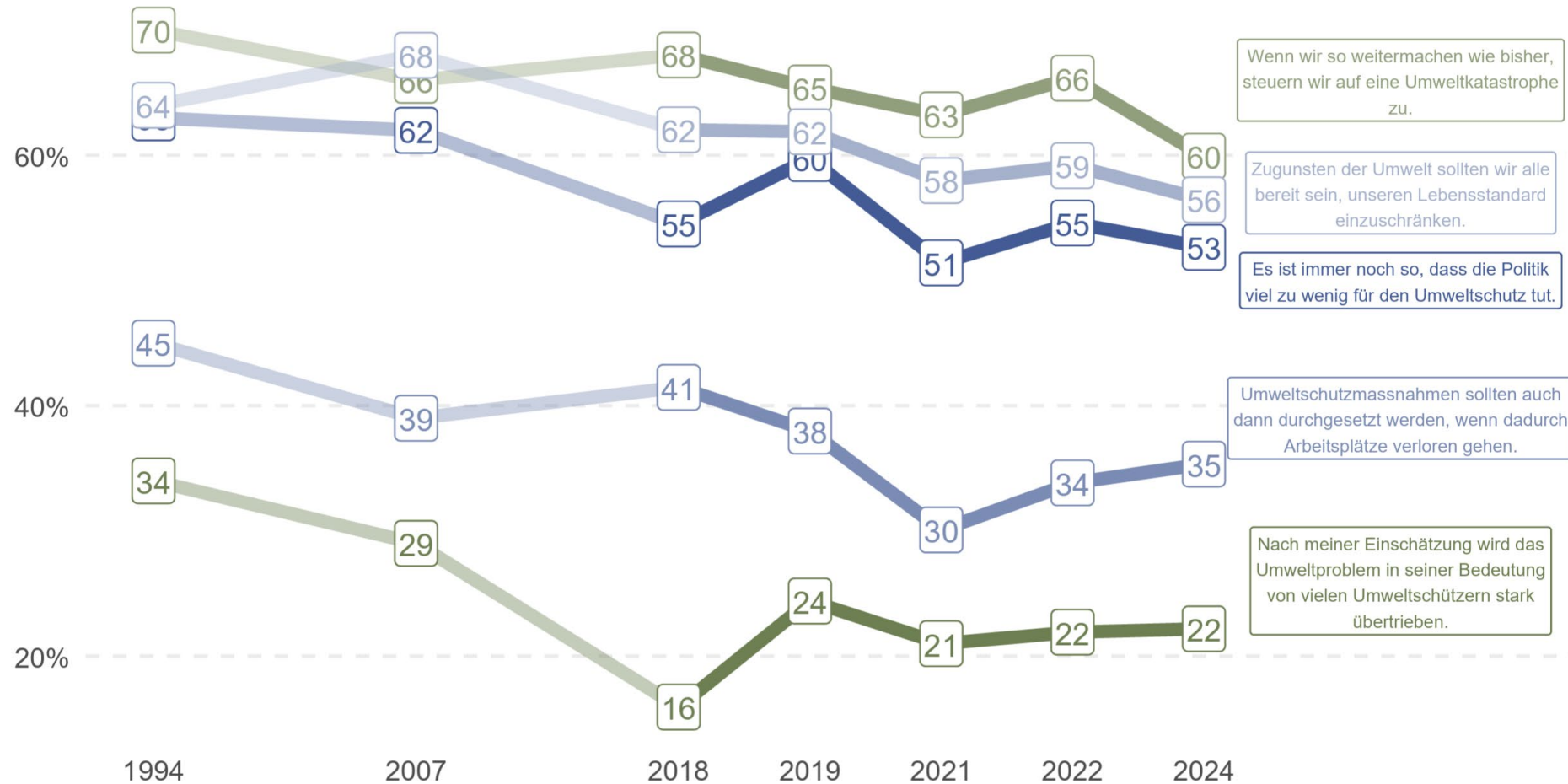
A climate of silence (N. Geiger & Swim, 2016)

- Inaccurate perceptions of others' opinions contribute to self-silencing among those concerned about climate change.
- Those who are aware of others' concern about climate change report greater willingness to discuss the issue than those with inaccurate perceptions of others' opinions.
- Correcting pluralistic ignorance increases concerned participants' willingness to discuss climate change.
- Cross-cultural study (11 countries) similarly finds small effects on increased willingness to express one's pro-climate opinion. (S. Geiger et al., 2025)

Public opinion about climate change – Switzerland

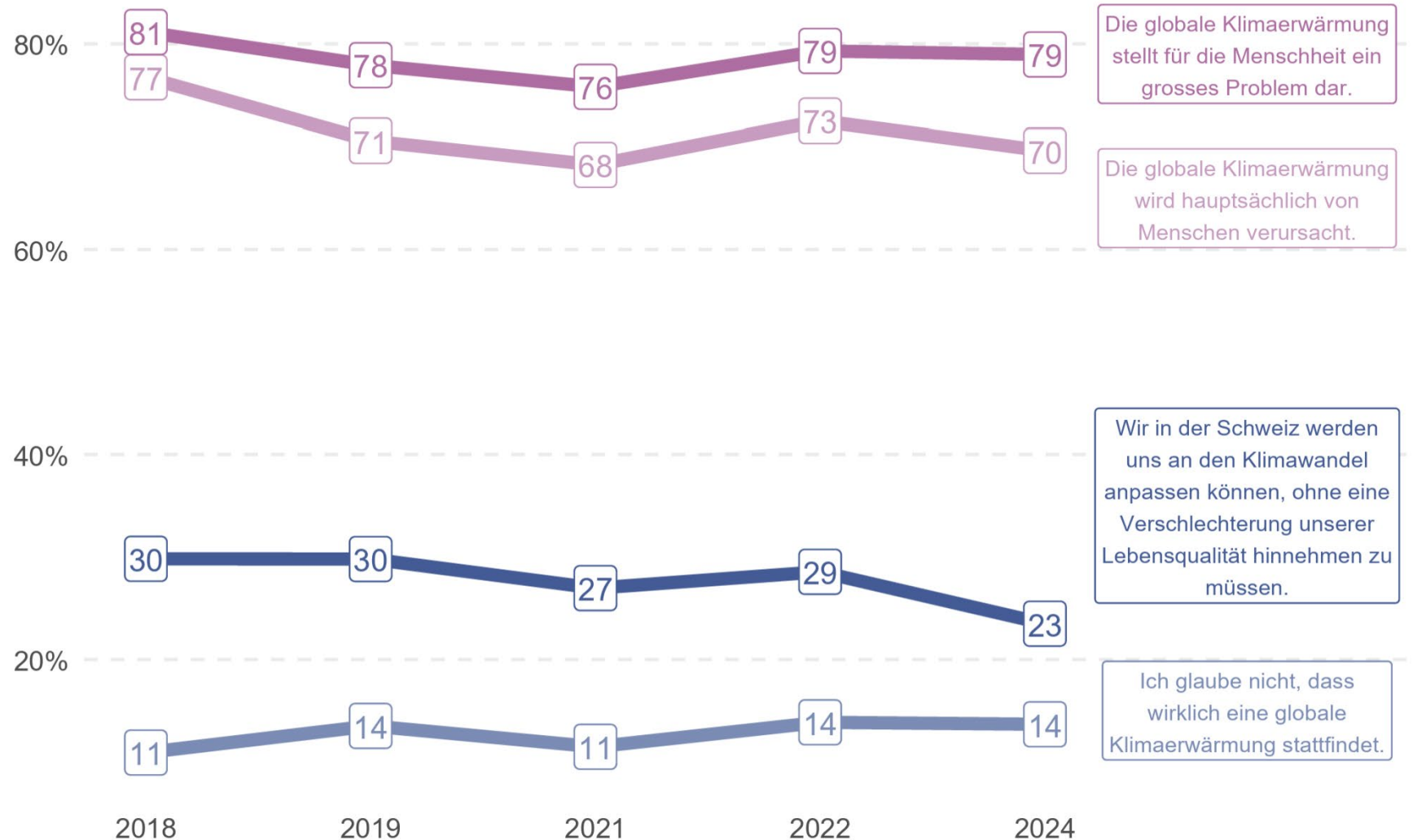
What about public opinion in Switzerland?

Public opinion about climate change – Switzerland



Fragetext: «Bitte geben Sie an, inwieweit Sie den folgenden Aussagen zustimmen oder nicht zustimmen.» Die Prozentzahl entspricht der Summe der Antwortmöglichkeiten «Stimme eher zu» und «Stimme voll zu». **Schweizer Umweltpanel, 2024**

Public opinion about climate change – Switzerland

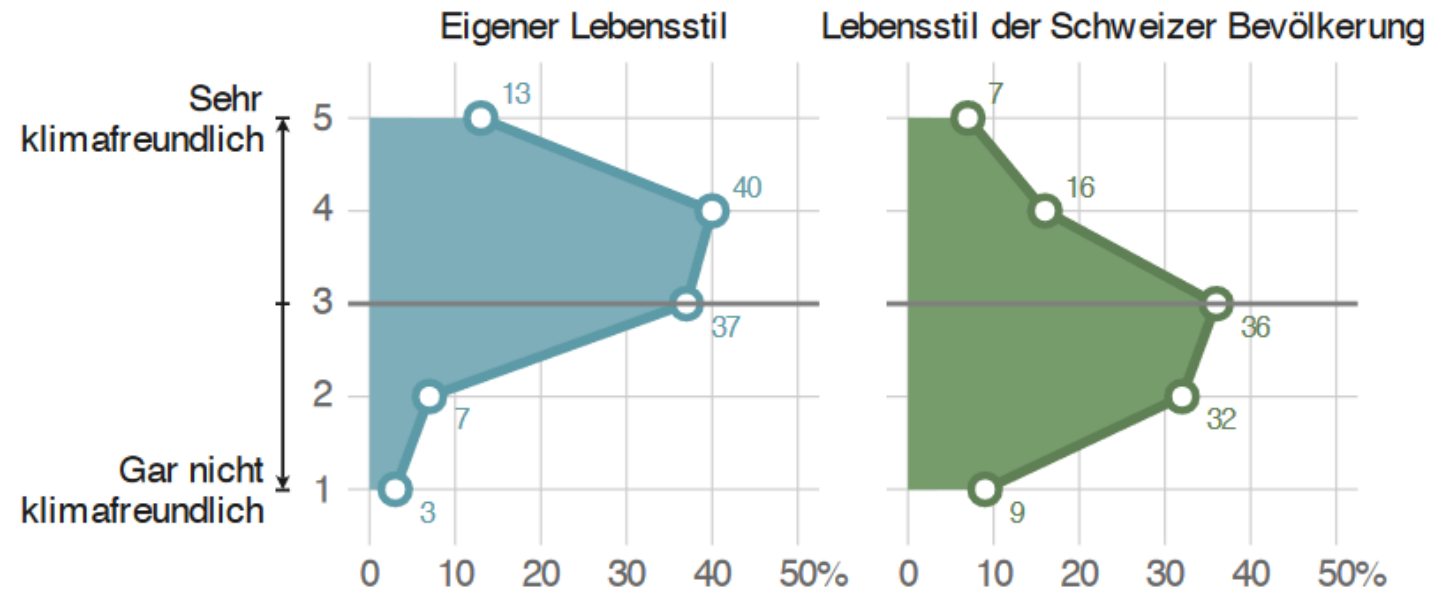


Schweizer Umweltpanel, 2024

Einschätzung der eigenen Klimafreundlichkeit im Vergleich zur Schweizer Bevölkerung (Abb. 13)

Lebensstil der Schweizer Bevölkerung: «Als wie klimafreundlich würden Sie den Lebensstil der Schweizer Bevölkerung im Durchschnitt einschätzen?»

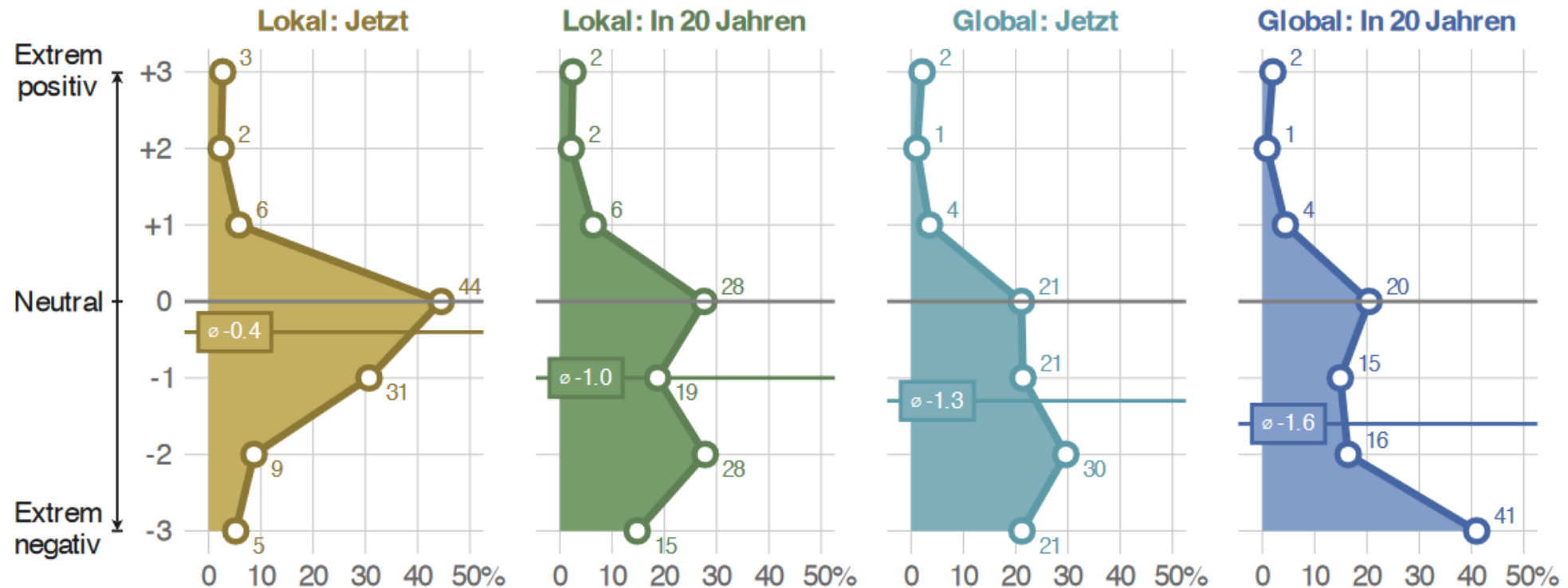
Eigener Lebensstil: «Als wie klimafreundlich würden Sie Ihren eigenen Lebensstil insgesamt einschätzen?»



Public opinion about climate change – Switzerland

Einschätzung der Auswirkungen des Klimawandels (Abb. 33)

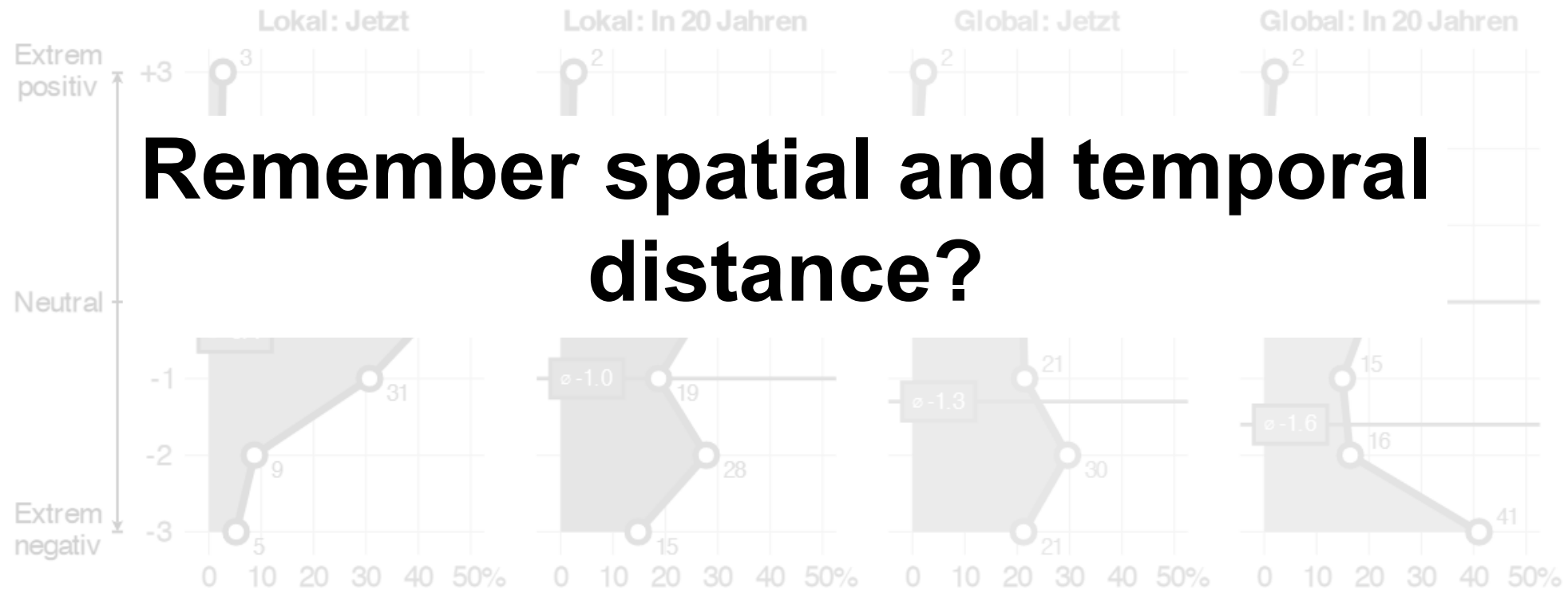
«Wie schätzen Sie die Auswirkungen des Klimawandels [zum jetzigen Zeitpunkt / in 20 Jahren] auf die Lebensumstände der Bevölkerung [an Ihrem Wohnort / in der gesamten Welt] ein?» (von -3: Extrem negativ, bis +3: Extrem positiv)



Helion Energiewende-Index (2024)

Einschätzung der Auswirkungen des Klimawandels (Abb. 33)

«Wie schätzen Sie die Auswirkungen des Klimawandels [zum jetzigen Zeitpunkt / in 20 Jahren] auf die Lebensumstände der Bevölkerung [an Ihrem Wohnort / in der gesamten Welt] ein?» (von -3: Extrem negativ, bis +3: Extrem positiv)



Let's discuss

- Turn to the person sitting next to you.
- In 2-3 minutes discuss the following questions:
 1. What was new/surprising to you?
 2. Why do you think concern about climate change is decreasing in Switzerland?

Why analyze public opinion on climate change?

- Track changes over time
 - and the effect of interventions
- Identify misperceptions of other's opinions
 - Prevent a climate of silence
 - Encourage pro-environmental behavior
- Develop effective communication and educational tools

Communicating climate change to different audiences

Article



Global warming's five Germanys: A typology of Germans' views on climate change and patterns of media use and information

Julia Metag, Tobias Füchslin and Mike S. Schäfer
University of Zurich, Switzerland

Public Understanding of Science
2017, Vol. 26(4) 434–451
© The Author(s) 2015
Reprints and permissions:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0963662515592558
journals.sagepub.com/home/pus



Julia Metag

Audience segments in Germany

Table 2. Audience segments and the climate change–related dimensions.

	Alarmed	Concerned activists	Cautious	Disengaged	Doubtful
N (% of sample)	459 (24%)	345 (18%)	543 (28%)	389 (20%)	201 (10%)
Concern about climate change	0.41	0.13	0.16	0.18	−1.60
Environmental concern	0.45	−0.02	−0.14	−0.19	0.33
Car use	0.05	−0.04	0.49	−0.39	0.26
Abstention from longer car/plane journeys	0.61	0.30	−0.65	0.45	−0.15
Eco-power	−0.42	1.75	−0.37	−0.34	−0.12
Political activism on energy issues	0.61	0.28	0.30	−0.93	−0.48
Ecological conservatism	−0.26	−0.24	0.26	0.01	0.11

N = 1937. Lowest and highest scores per factor are shown in bold.

Metag et al., 2017

Communicating climate change to different audiences

- Different audience types differ in their media use and information-seeking patterns.
- Helps to understand how to address different audiences in climate change communication

Communicating climate change to different audiences

- **Disengaged:** They are not well-informed. Communication campaigns should be easily understandable and stress inexpensive methods for changing behavior. Since the Disengaged use tabloids, they might respond better to entertainment or fictional formats, such as movies.
- **Doubtful:** A campaign on global warming should be aimed at **raising their concern** about climate change and reducing their doubts about ongoing global warming.
- **Cautious:** The Cautious are aware of climate change as a problem but must be motivated to change their behavior accordingly. Communication campaigns should focus on **mobilizing information** and provide information about behavioral options in everyday life.
- **Alarmed:** Campaign should focus on **inexpensive ways to behave environmentally friendly** since this group is hesitant to use more expensive methods. These people could potentially function as opinion leaders as they talk about climate change quite often. A campaign should provide information about climate change that is easy to communicate further to other people.
- **Concerned Activists:** Young and more affluent. These people are concerned about climate change but not as much about the environment in general. Information should be provided not only on television and the Internet but also in print media (note 2016).

Communicating climate change to different audiences

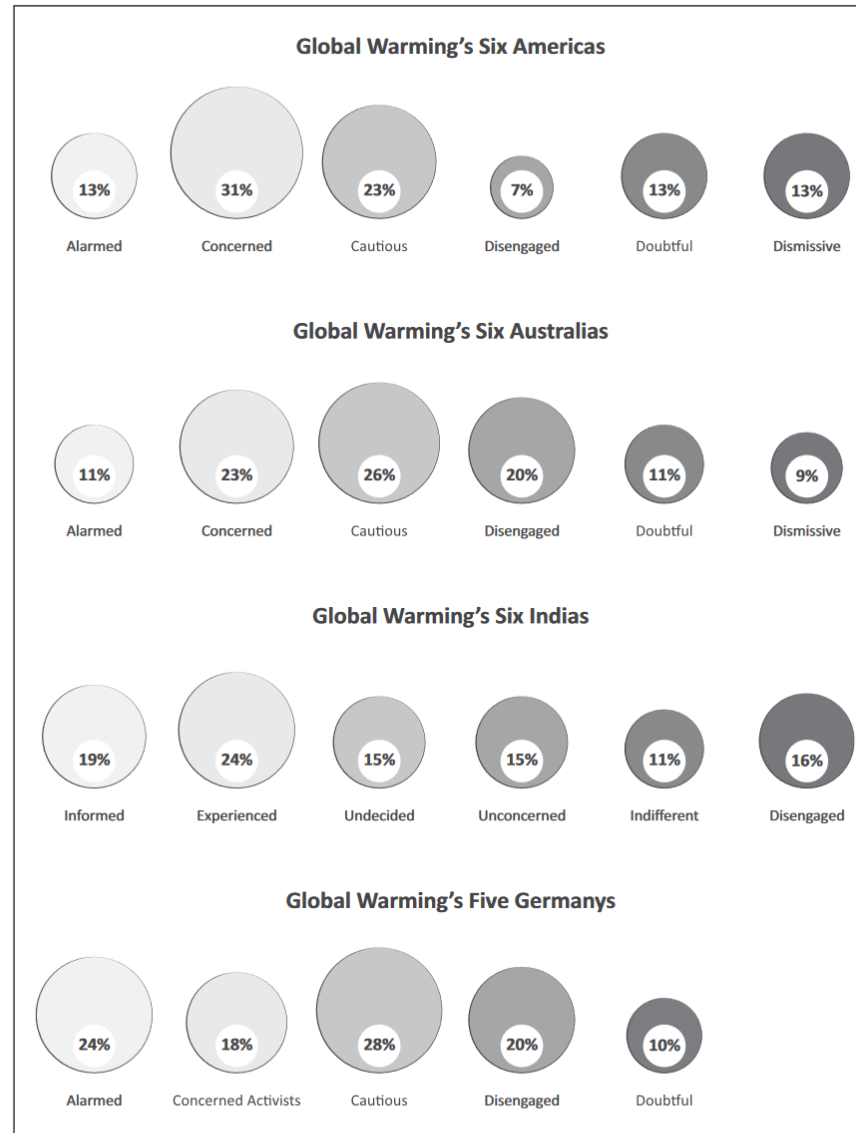
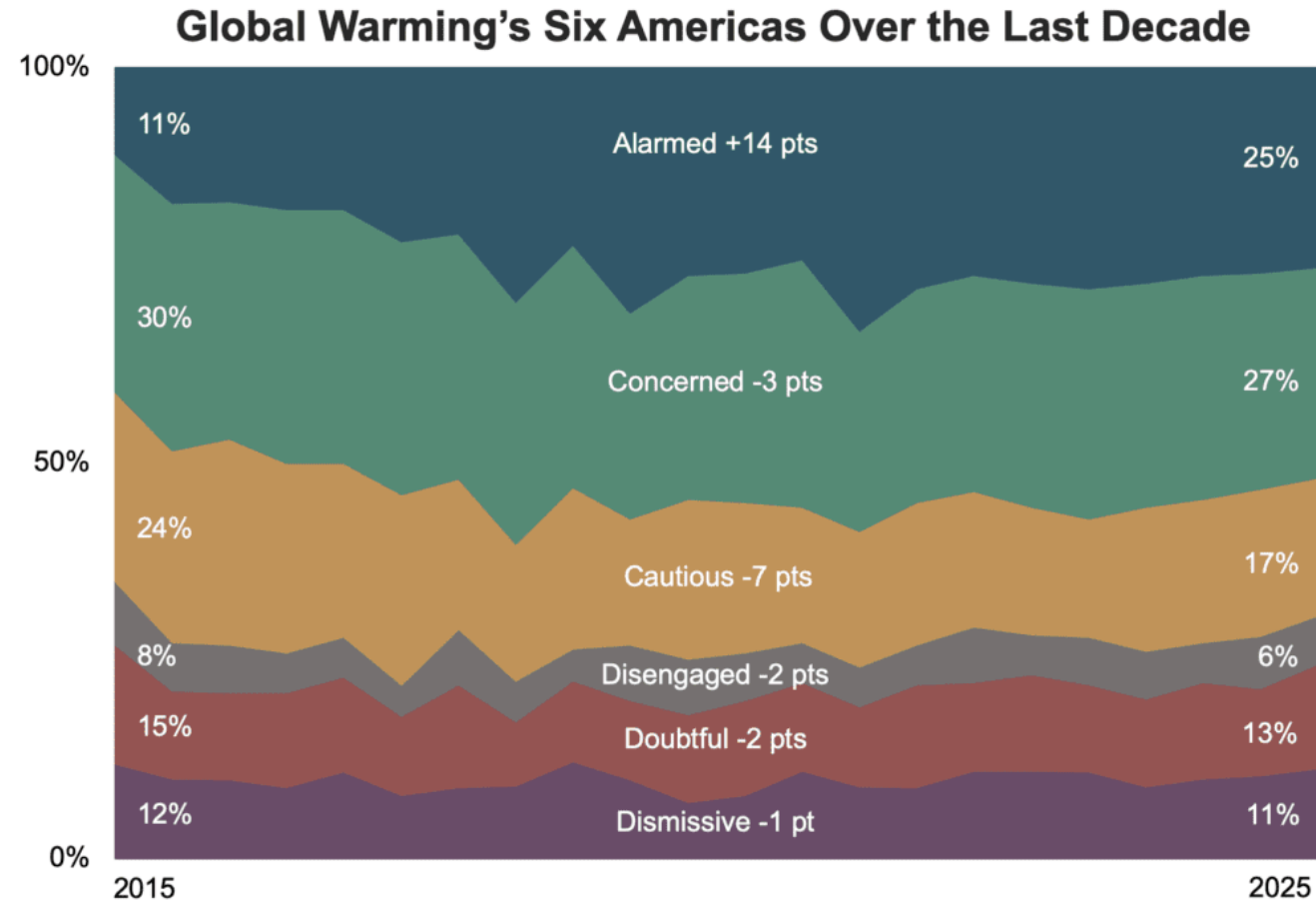


Figure 2. Audience segments in different countries (data are retrieved from Leiserowitz et al. (2013b, 2014) and Morrison et al. (2013)). Comparability of the segments is limited as there are differences in survey items and data analysis between the countries.

Communicating climate change to different audiences



Base: 25,064 U.S. adults. Data include 22 waves of national surveys spanning March 2015 – November 2025.

Source: Yale Program on Climate Change Communication;
George Mason University Center for Climate Change Communication

Do we need to communicate more facts?

- 60% of people in the USA and 70% of people in Switzerland believe that climate change is human-caused.
- **Deficit model of science communication:** “assumes that increased communication and awareness about scientific issues will move public opinion toward the scientific consensus and reduce political polarization around science-based policy” (Hart & Nisbet, 2012, pp. 701-702).

The deficit model

Do we need to communicate more facts?



Lawrence Livermore National Laboratory

More knowledge about climate facts = more concern?

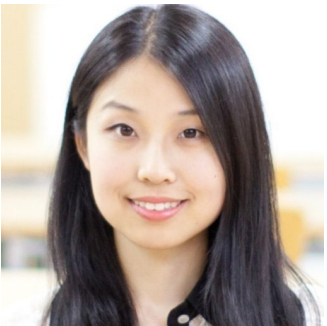
Letter | Published: 25 April 2016

Knowledge as a driver of public perceptions about climate change reassessed

[Jing Shi](#) , [Vivianne H. M. Visschers](#), [Michael Siegrist](#) & [Joseph Arvai](#)

[Nature Climate Change](#) **6**, 759–762 (2016) | [Cite this article](#)

9440 Accesses | **198** Citations | **309** Altmetric | [Metrics](#)



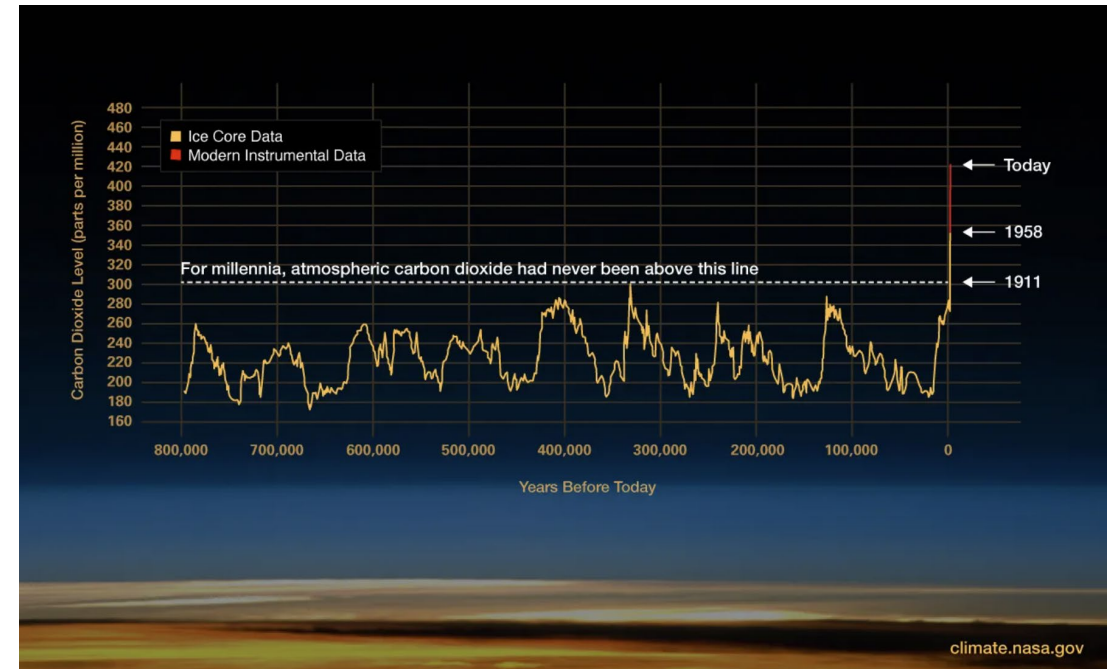
Jing Shi

“Recent research suggests that knowledge about climate change has only a limited effect on shaping concern about climate change. **Our view is that this counterintuitive finding is a function of how knowledge is typically measured in studies about climate change.**”

What type of facts should be communicated?

Different types of climate change knowledge:

- Knowledge about physical characteristics



What type of facts should be communicated?

Different types of climate change knowledge:

- Knowledge about physical characteristics
- Knowledge about causes



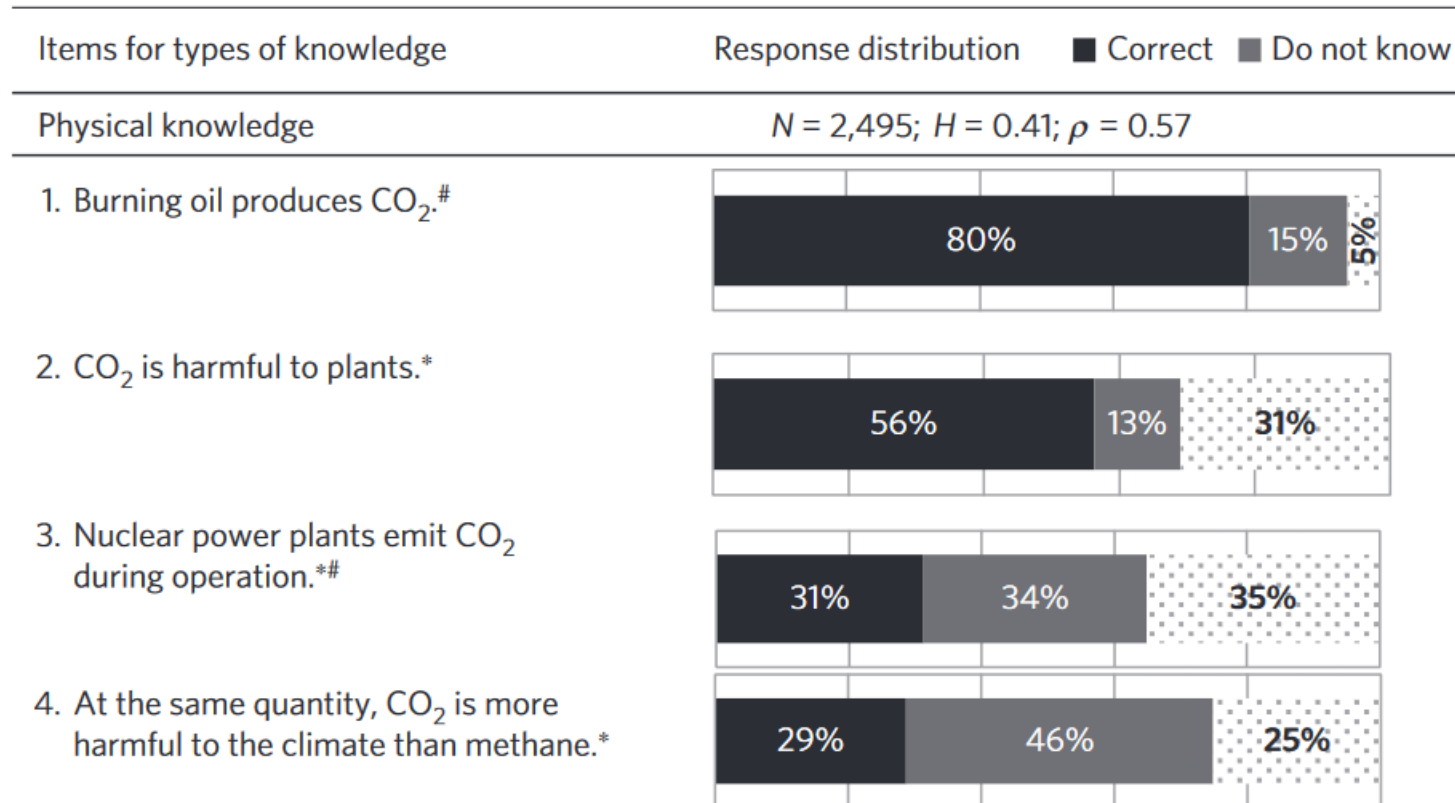
What type of facts should be communicated?

Different types of climate change knowledge:

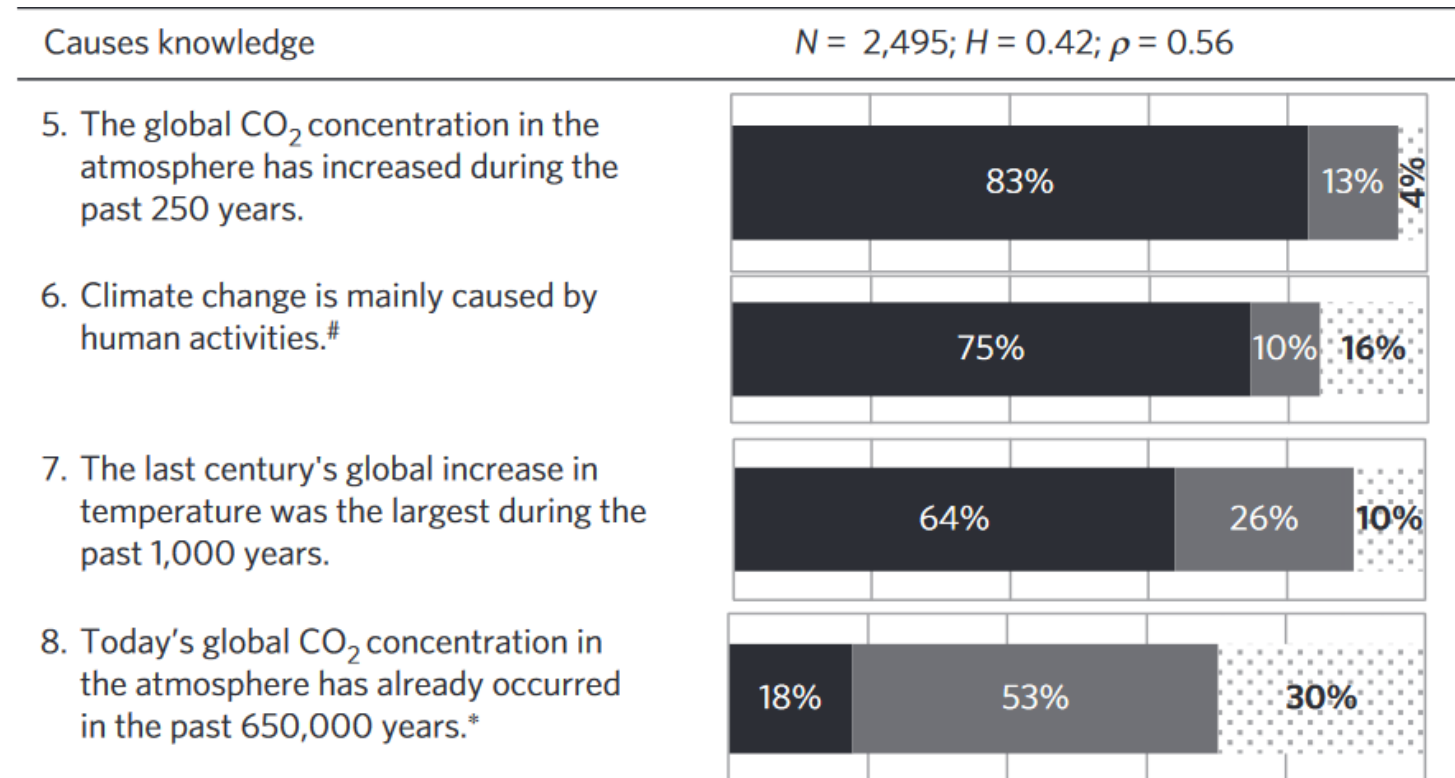
- Knowledge about physical characteristics
- Knowledge about causes
- Knowledge about consequences



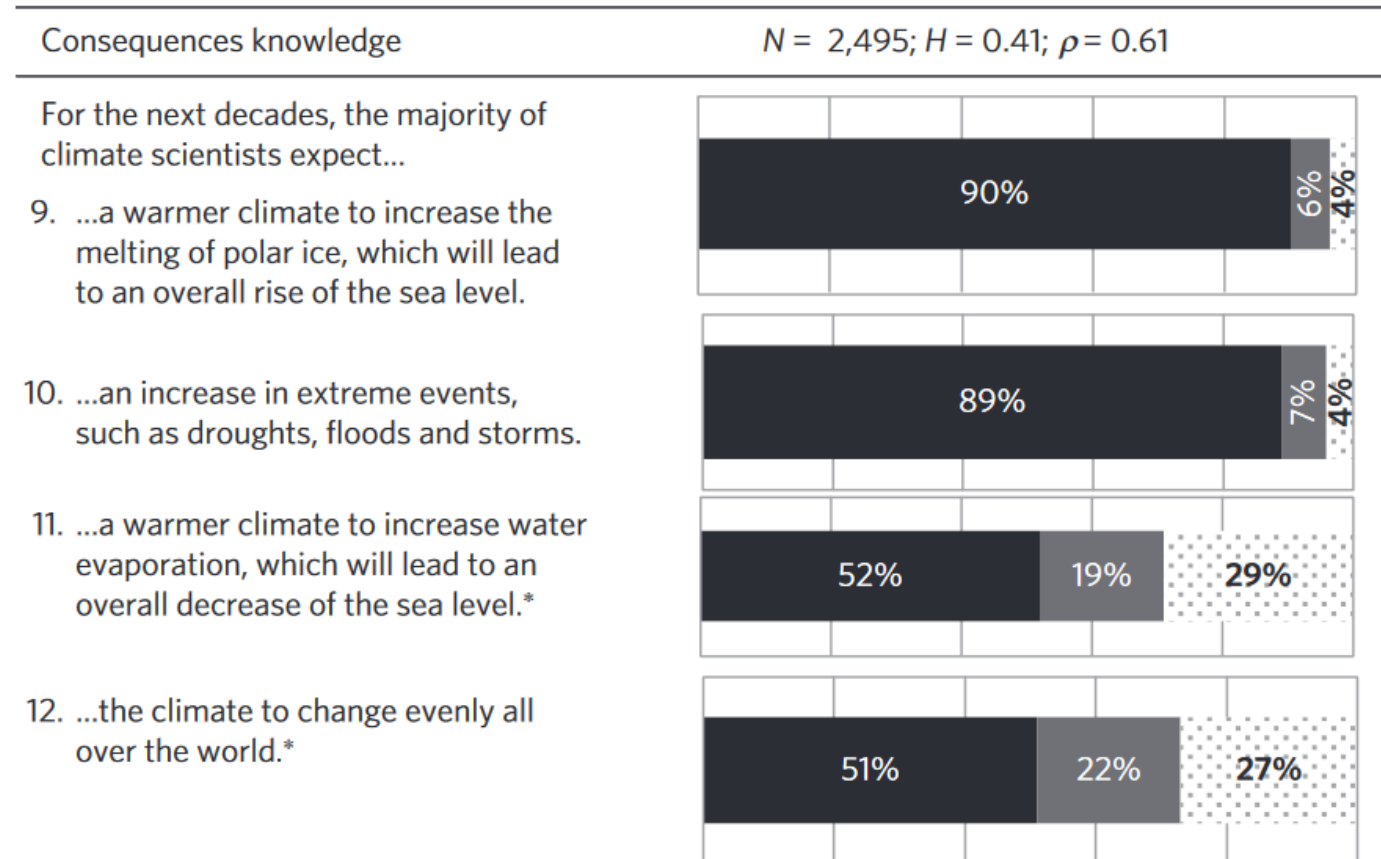
Knowledge on physical characteristics



Knowledge on causes



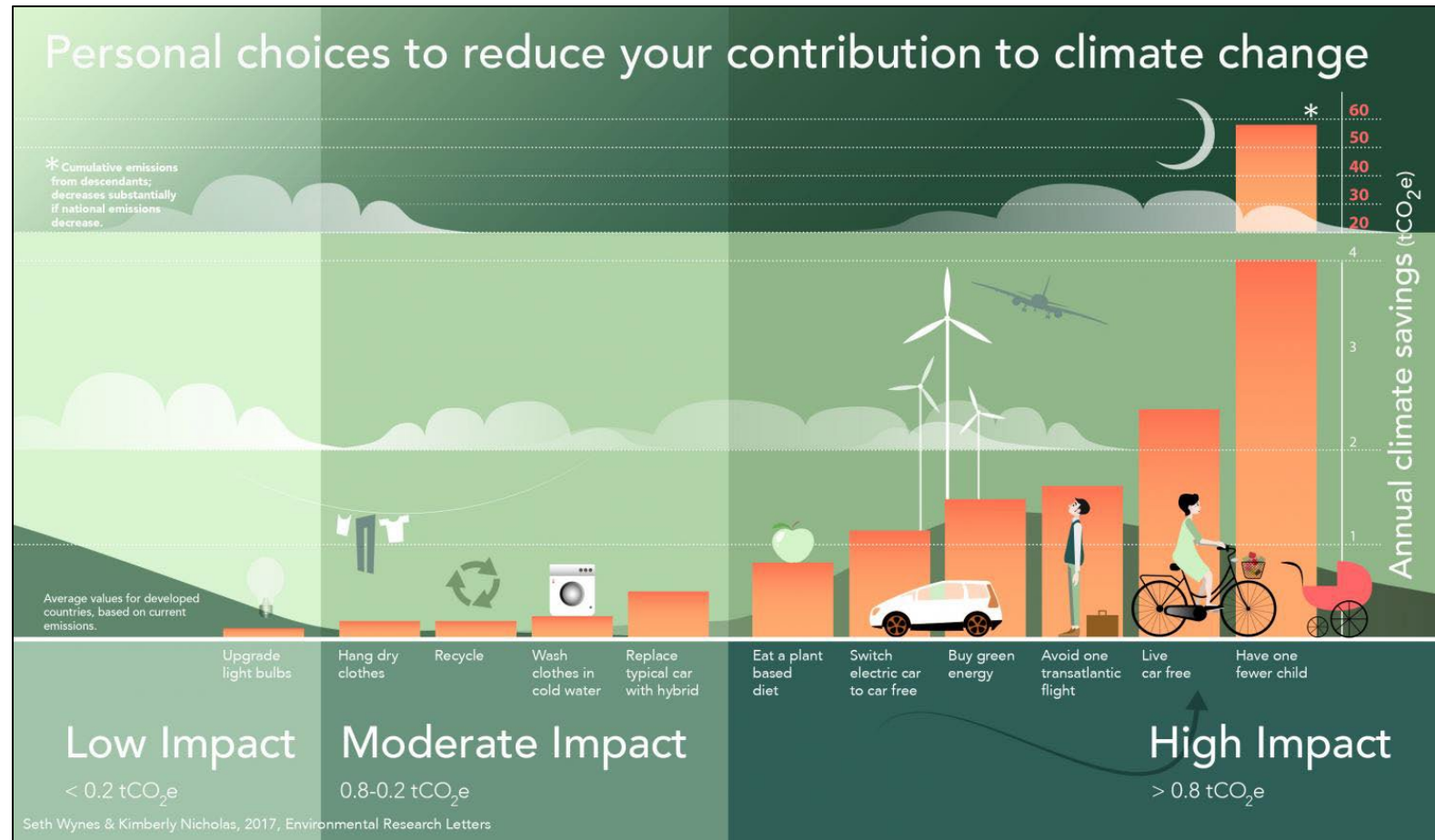
Knowledge on consequences



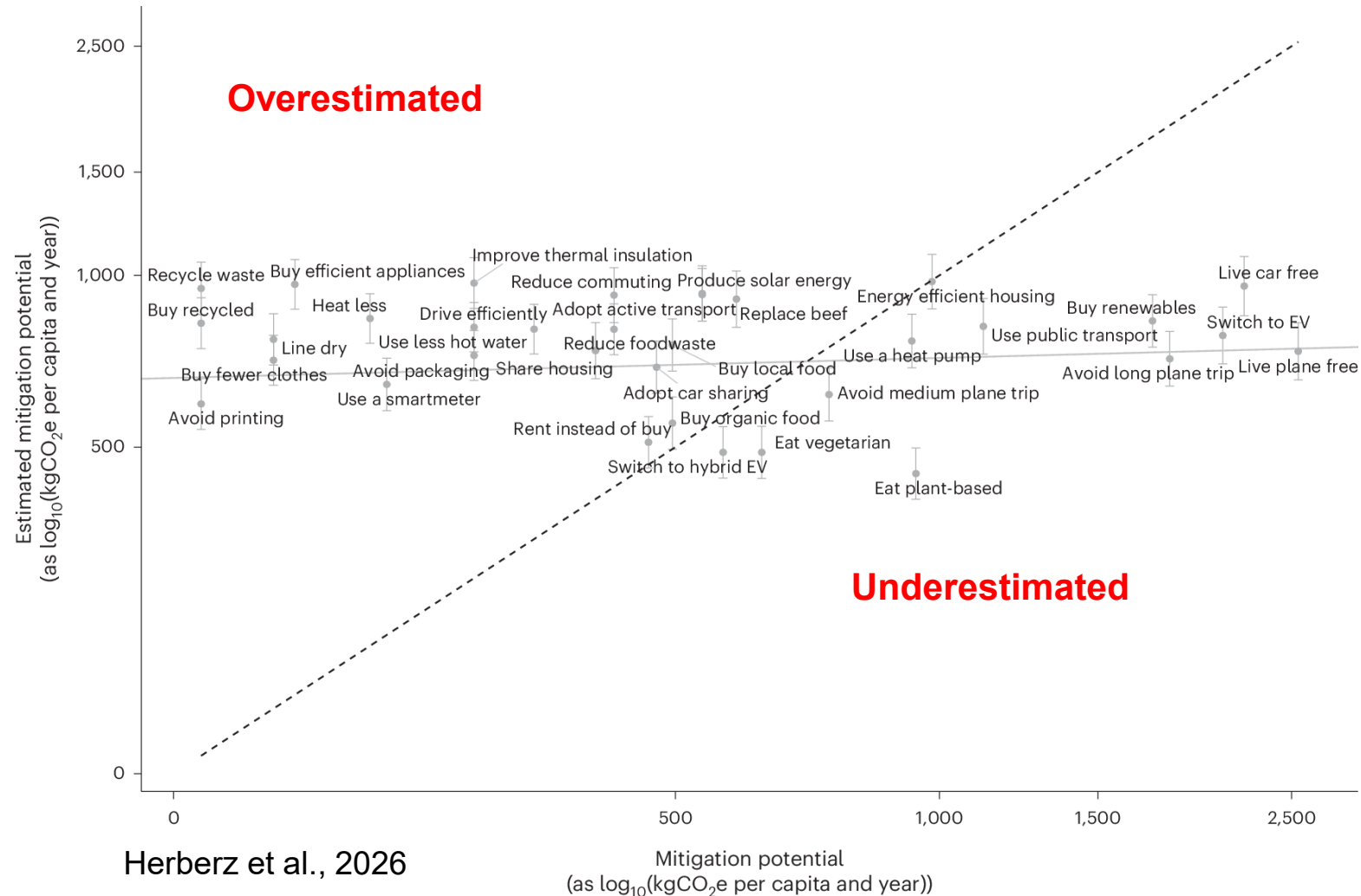
What type of facts should be communicated?

- Knowledge about **physical characteristics**: tended to dampen public concern about climate change
- Knowledge about **causes**: higher levels of climate change concern
- Knowledge about **consequences**: higher levels of climate change concern

What about communicating the impact of different behaviors?



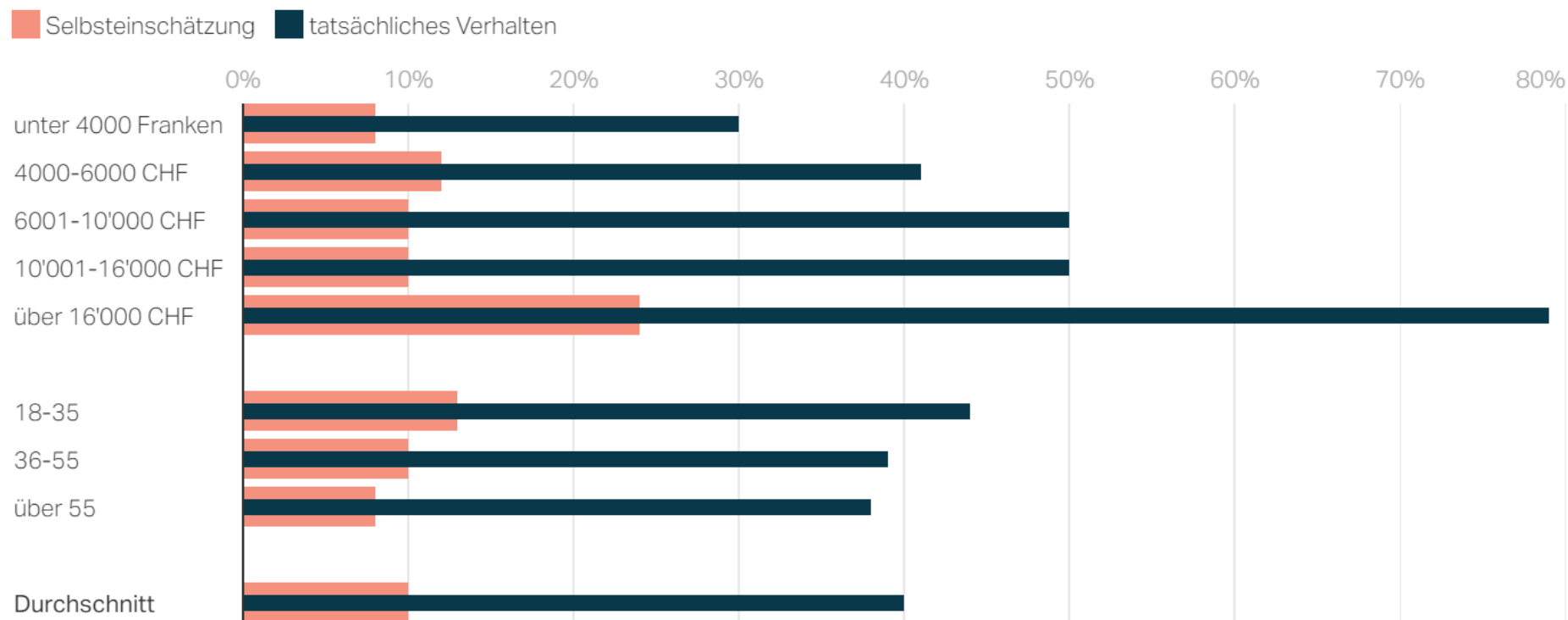
Misperceptions about mitigation potential of behaviors



Same holds true in Switzerland

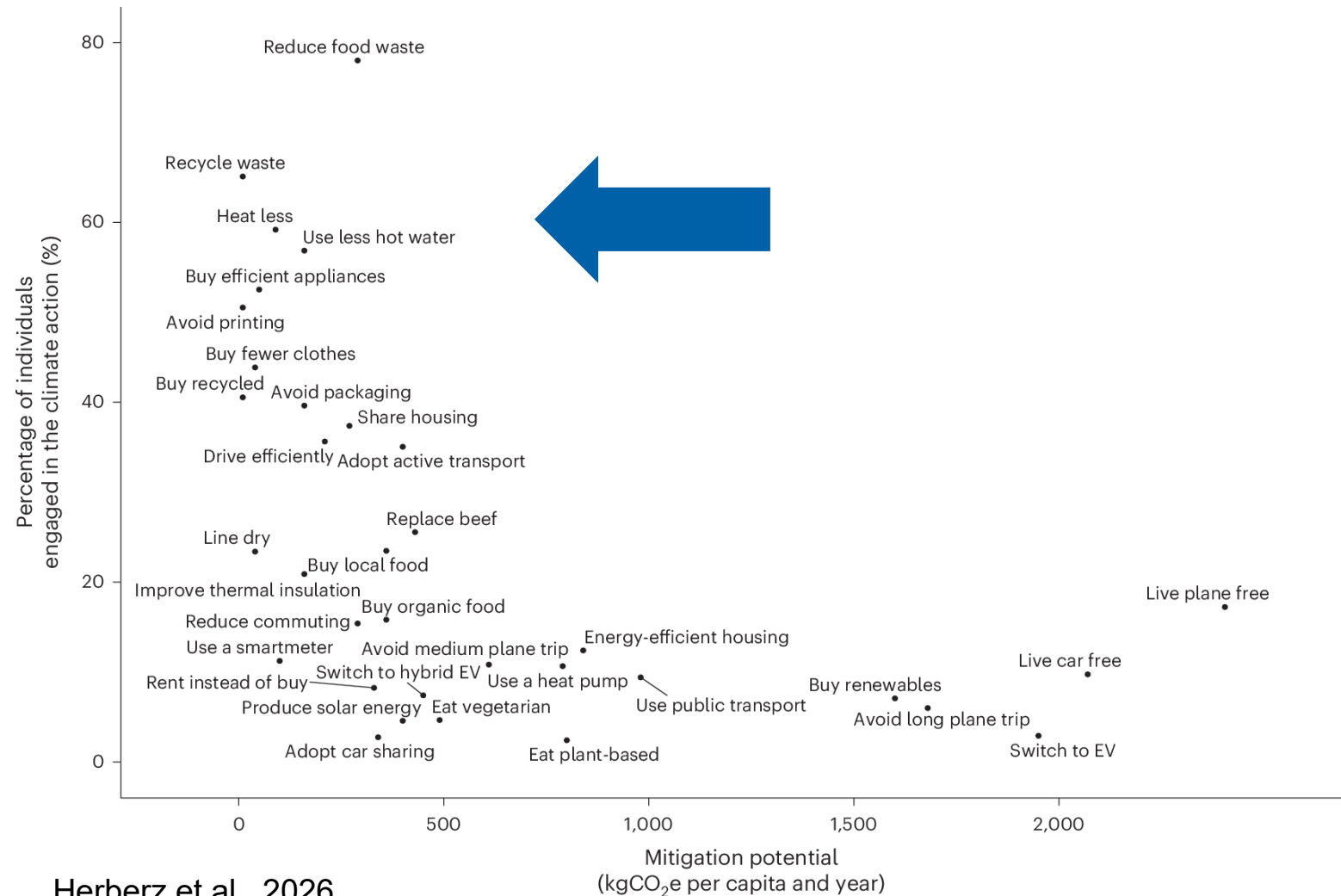
Die meisten Schweizerinnen und Schweizer unterschätzen die Grösse ihres ökologischen Fussabdrucks

Frage: Ist mein CO₂-Fussabdruck überdurchschnittlich? Selbsteinschätzung und Realität



So glauben 24% der Schweizer, die mehr als 16'000 CHF im Monat verdienen, dass ihre jährlichen CO₂-Emissionen überdurchschnittlich hoch sind. In Wirklichkeit haben 79% der Menschen in dieser Einkommensklasse überdurchschnittliche jährliche CO₂-Emissionen.

People mostly engage in low-impact climate behaviors



What explains this lack of carbon competence?

- **Availability heuristic:** people judge an action as more effective if it easily comes to mind
- **Motivated reasoning:** serves to preserve a positive self-image (Herberz et al., 2026)

→ This can lead to people misdirecting their efforts!

Correcting misperceptions

- Matters because people with more knowledge about carbon footprints report lower carbon footprints themselves (Frings et al., 2025)
- Providing information about the actual mitigation potential of behaviors can correct misperceptions and increase commitments toward more effective individual-level climate actions (Goldwert et al., 2025)

Correcting misperceptions

→ However, this can backfire on commitments to engage in collective action

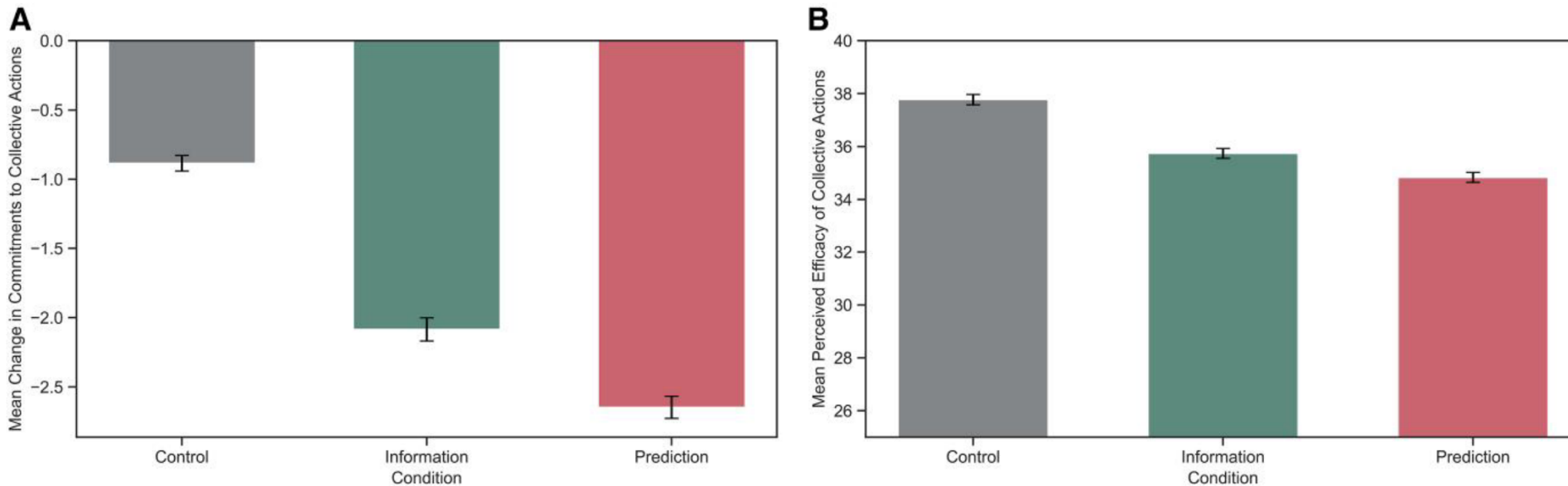


Fig. 5. A) Changes in commitments from pretest to posttest in collective actions as a function of experimental condition. B) Changes in perceptions of efficacy of collective actions as a function of experimental condition. Error bars represent SEs of the mean.

Let's wrap up

- Public opinion data help communicators understand the size of different audiences
- People underestimate the scientific consensus on climate change and other people's support for climate action (pluralistic ignorance)
 - Communicating about the scientific consensus can increase belief and worry
- The importance of knowledge for climate action depends on the type of knowledge
 - Communicate facts about the causes and consequences
- People systematically underestimate the impact of high-impact behaviors
 - Communication on behaviors' impact can increase engagement in these behaviors, but potentially backfire on collective action